

職業性傳染病

Occupational Infectious Disease



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What is a biological hazard?

- Biological hazards, also known as **biohazards**, refer to biological substances that pose a threat to the health of living organisms, primarily humans.

Classifications

p **Microorganisms**

- ✓ Examples: Virus, bacteria, rickettsia, fungi, and parasites
- ✓ Reaction: Infection

p **Allergens of biological origin**

- ✓ Examples: aeroallergenic fungi, animal dander, pollens
- ✓ Reaction: Allergy

p **By-products of microbial growth**

- ✓ Examples: endotoxins, exotoxins, mycotoxins, MVOCs (Microbial Volatile Organic Compounds)
- ✓ Reaction: toxic effect

Routes of transmission

Direct

- ❖ Skin-skin
 - ❖ Herpes type 1
- ❖ Mucous-mucous
 - ❖ STI
- ❖ Blood, transplants
- ❖ Across placenta
 - ❖ toxoplasmosis
- ❖ Sneeze-cough (droplet)
 - ❖ Influenza

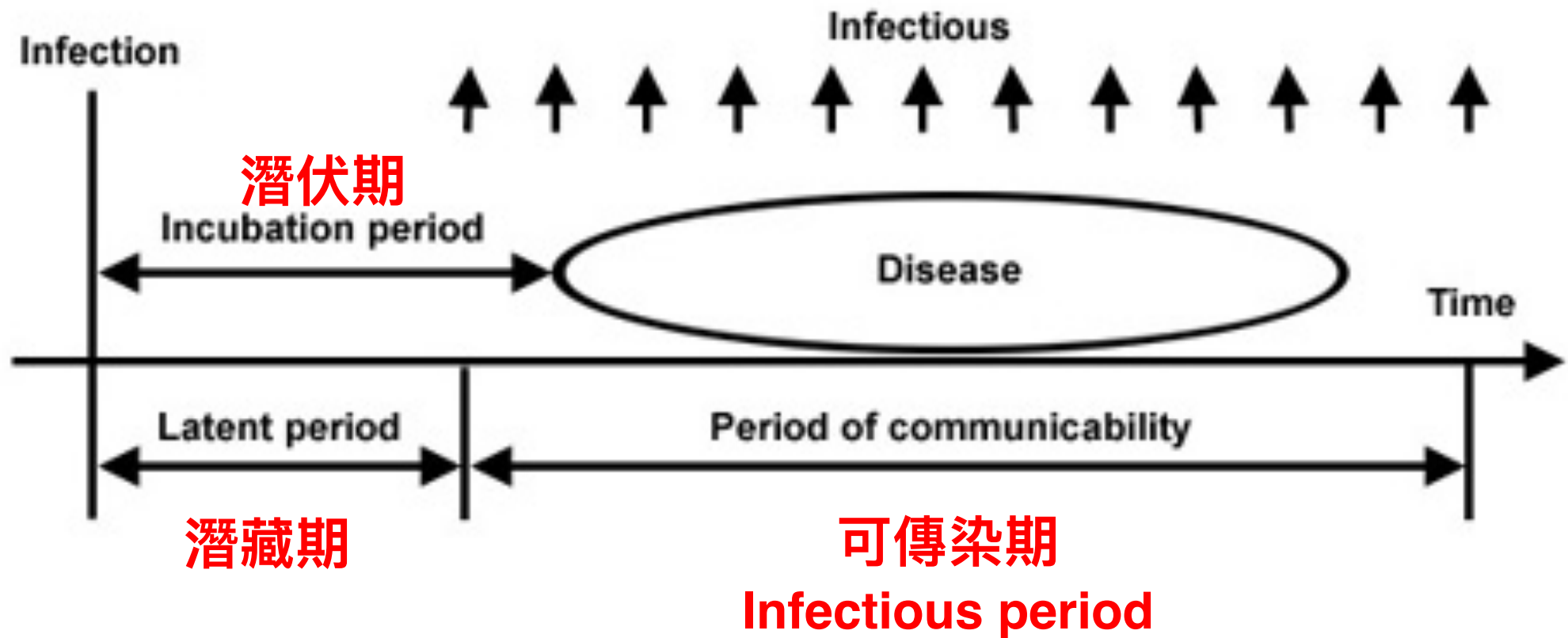
Indirect

- ❖ Food-borne 傳播
 - ❖ Salmonella
- ❖ Water-borne
 - ❖ Hepatitis A
- ❖ Vector-borne 蟲媒傳染
 - ❖ Malaria
- ❖ Air-borne
 - ❖ Chickenpox
- ❖ Objects and fomites

Exposure

- ❖ A relevant contact – depends on the agent
 - ❖ Skin, sexual intercourse, water contact, etc

Incubation and latent period



Infectivity

- The ability of an agent to cause infection in a susceptible host.
- Basic measure:
 - ✓ minimum infectious dose
 - ✓ median infectious dose
- In diseases spread from **person to person**, the proportion of susceptible individuals who develop infection after exposure - **the secondary attack** rate - is a measure of the infectivity of an organism. 指標

Infectious Dose Estimates

Pathogen	ID ₅₀
Shigella dysenteriae	$\sim 10^2$
V. cholerae	$< 10^6$
Salmonella typhi	$\sim 10^7$
Nontyphoid Salmonella	$\sim 10^5$
Pathogenic E. coli (several strains)	$\sim 10^7$
E. coli O157:H7	"low"
Rotavirus 諾羅	5-6 ffu
Norwalk virus	430 PDU
Cryptosporidium	132-238 oocysts

Pathogenicity

- *The ability of a microbial agent to induce disease.* ↳ 病使宿主生病的能力
- High pathogenicity: rabies, smallpox, measles, chickenpox, and rhinovirus colds
- Low pathogenicity: polio and arbovirus (mosquito-borne) infections

Percent of Infected with Clinical Symptoms

- Poliovirus: 0.1-1% (children)
- Hepatitis A: 5% (children), 50-75% (adult)
- Norovirus: 83-88% (challenge study data)
- Giardia: ~50% (outbreak data)
- Salmonella (Nontyphi): 41% (outbreak data)
- V. Cholerae: Dose-dependent

Virulence

引起重症的能力

- Some dictionaries use the terms *virulence* and *pathogenicity* interchangeably
- *Virulence can be defined as the severity of the disease after infection occurs*
- Virulence can best be measured by the case fatality rate or as the proportion of clinical cases that develop severe disease.

Classification of Several Infections Diseases by their Epidemiologic Characteristics

Table	6	Classification of Several Infections Diseases by their Epidemiologic Characteristics		
Severity	Infectivity	Pathogenicity	Virulence	
High:	(Secondary AR) Measles Pertussis Ebola HPV SARS/ COV Influenza	(Illness=Infection) Measles Rabies Pertussis Ebola Smallpox Rhinovirus-URI SARS/COV Influenza	(Severe/Fatal) Rabies Ebola SARS/COV Smallpox	
Intermediate	Smallpox Rubella Mumps Polio Rhinovirus URI	Rubella Mumps	Polio Pertussis Measles Tuberculosis Leprosy Influenza	
Low	Tuberculosis Leprosy	Polio Tuberculosis Leprosy	Rhinovirus-URI Rubella Mumps	

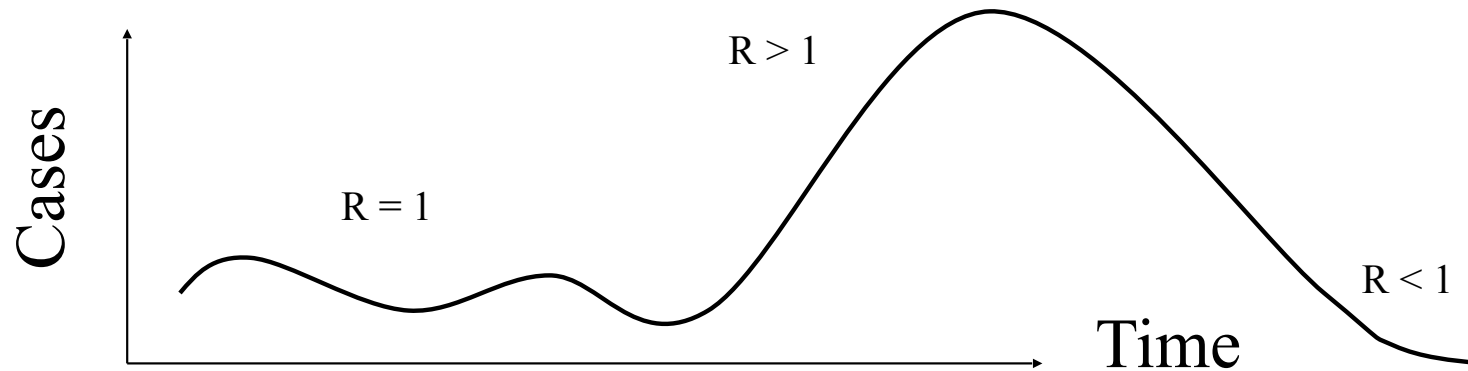
Isolation and Quarantine

- **Quarantine (檢疫)** involves the separation or restriction of movement of well individuals who may have been exposed to an infectious disease to see if they become sick. ↳ 未發病
 - Duration depends on testing and incubation period
- **Isolation (隔離)** involves the separation of infected and sick individuals from healthy individuals. ↳ 已發病
 - Duration depends on treatment and infectious period

Reproductive number

- **Basic reproductive number (R_0 , 基礎再生數) :**
 - The average number of individuals directly infected by an infectious case during his or her entire infectious period, when he or she enters a **totally susceptible population**.
- **Effective reproductive number (R_E , 有效再生數)**
 - $R > 1 \rightarrow$ 疫情會惡化造成流行
 - $R = 1$ ☐ 變成地方性流行
 - $R < 1 \rightarrow$ 疫情會減緩而消滅

Endemic - Epidemic - Pandemic



❖ Endemic

- ❖ Transmission occurs, but the number of cases remains constant

❖ Epidemic

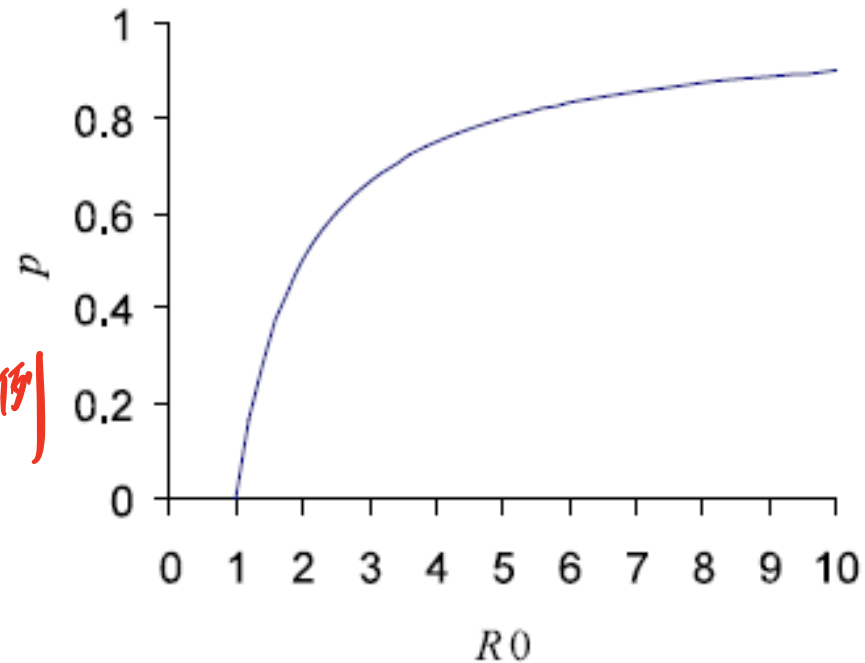
- ❖ The occurrence of a disease or condition at a frequency that is unusual or unexpected
 - Increase above background or endemic level

Vaccination

Proportion of the population which need to be effectively protected against infection via vaccination at birth to eliminate a disease:

$$p = 1 - \frac{1}{R_0}$$

“需打疫苗才能免疫的比例”



... though this simple result does depend on random mixing.

Heterogeneity in contact patterns can change everything!

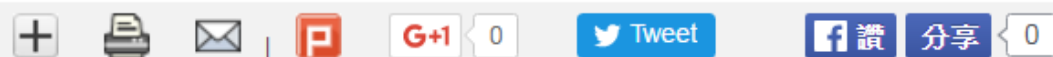
- Hence control policies need to eliminate a fraction $1 - 1/R_0$ of transmission – i.e. 50% for $R_0 = 2$, 75% for $R_0 = 4$, 90% for $R_0 = 10$.

COVID-19 pandemic

WHO estimates that between 80 000 and 180 000 health and care workers could have died from COVID-19 in the period between January 2020 to May 2021, converging to a medium scenario of 115 500 deaths¹.



洛杉磯立法拍片須帶保險套 引發A片大廠反彈



2012-01-25 12:59

〔即時新聞／綜合報導〕媒體報導，美國加州洛杉磯是著名的色情電影產地，但在去年陸續傳出有AV女星感染愛滋病後，當局日前通過草案，規定所有A片女星在拍片前都要做好保護措施，強迫規定必須戴上保險套。此舉雖然是出於美意，但卻是引發不少A片廠商及AV演員的強烈反彈。

據BBC報導，美國AV演員未來在加州洛杉磯拍攝色情影片時，都得遵照規定使用保險套，而且新法案在星期一已由洛杉磯市長簽署。新規定的上路雖然大受衛生官員的歡迎，卻引發不少A片大廠及A片名星的排斥。

A片工業領導人甚至出面表示，他們將要拋棄洛杉磯，尋找其他能合法不使用保險套拍片的城市。然而據特派員表示，市政府尚未確定新規定將如何施行。

此法案的實施令洛杉磯愛滋病防治基金會相當地振奮，因為他們努力奔走並且宣傳長達了將近6年的時間，直到近日才開花結果，而他們也表示，他們將要尋找下一個需要被立法保護的城市，繼續來進行他們的宣導立法動。

對此，則是有許多A片大廠紛紛跳出來痛斥，他們將要遠離洛杉磯另起爐灶，而且美國有90%的A片是來自於洛杉磯，此法上路後將會大大衝擊該地的產業。更有A片廠商強調表示，使用保險套不但影響他們的銷量，甚至有些是A片演員自己傾向不使用保險套來拍片的。

▲加州擬訂演員在拍A片的時候要戴上護目鏡，引起業者反彈。（圖／翻攝自《華盛頓郵報》）

國際中心／綜合報導

加州職業安全與健康標準（[OSHA](#)）上週提出演員在拍A片的時候，不僅要戴上保險套，還要強制必須使用護目鏡，A片業者跳出來說，這個新規則是沒有必要的。

據《華盛頓郵報》報導，愛滋病保健基金會長[Michael Weinstein](#)表示，新規則是為了保護A片演員，因為成人電影行業裡至少有4人已經感染愛滋病（[HIV](#)），還有千上萬的人染上其他的性傳播疾病。

據報導，成人娛樂行業的貿易協會的首席執行官[Diane Duke](#)說，加州是美國A片的大產地，洛杉磯在2012年時通過一個爭議的法條，就是演員在拍攝時一定要戴上保險套，結果生產率下降了9成以上，製作公司也幾乎轉到拉斯維加斯和南佛羅里達州拍攝。她說，這是醫療環境的法規，在成人片、好萊塢電影集上是行不通的，將這些規則套用在表演上的話，會影響整個A片產業。

20160926 15:00
這區房子7成老摳
摳 健檢到著等
(3899)

20160926 14:57
醉乘客國道瑞門
吵下車 全車乘
客都被延誤(526)

20160926 14:57
斷續漂流的風明輪
今拖回船塢防颱(0)

20160926 14:56
謝國樑宣佈退出競選
刊經營 因「理念路
線差異」(0)

20160926 14:55
【也是亂象】梅姬明
來襲 立院竟要立委
先放棄質詢(5921)

20160926 14:55
父po文嘆11歲自閉兒
沒朋友 結果令人心
暖(177)

20160926 14:54
【不斷更新】梅姬擾
局 全台活動喊停看
這裡(7622)

20160926 14:53
飛行途中機艙冒煙
菲航飛日班機折返馬
尼拉(32)

20160926 14:52
【有片】蘋概股
領跌 台股跌90
點失守9200點
(948)

20160926 14:52
多元應用蓬勃發展
紅外線LED市場大爆
發(13)

f 分享FB g+ 分享g+ Plurk 分享Plurk Twitter 分享Twitter

2016年02月22日23:20 傳送 1 G+ 0

[原文影片請看此](#)

加州職業安全健康局官員否決了A片演員應戴保險套，拍攝性愛場面的提案。

週四一項表決，以3比2的票數，當局官員與色情片業者站在同一陣線，否決了這項提案。

AV女優Maxine Holloway向職業安全與健康標準委員會（Division of Occupational Safety and Health's Standards Board）懇求道：「拜託別通過這項政策，因為這會危及我和同事們的健康。」

反對戴套的一種說法是，觀眾不喜歡看演員戴套的A片。

許多色情片業界人士聲稱，若這項規定實施，他們就要離開加州。

洛杉磯是產值數十億美元的色情影業重鎮，目前在洛城，A片演員依規定要戴套拍片，雖然很多人忽視這項規定。

反對戴套的人士說，若規定要戴保險套拍片，那以後可能會有更極端的法案，規定演員在鏡頭前要戴蛙鏡、手套、或牙醫橡皮障。

公聽會長達5小時，到場的色情影業人士聽到表決結果後，大聲歡呼。

當然，不愛看戴套鹹濕片的觀眾也同聲歡呼囉。

英文原文：

財經最 Hot

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蘋果粉絲團



蘋果日報

說這專頁讚 331 萬 按讚次數



蘋果日報
2 mins

「學心理的人切記：跨過紅線，大師也會變人渣。」

【#火線話題】輔大性侵爭議

<https://goo.gl/mLppBF>

..... 更多

翻譯年糕



虎航空姐得麻疹 同機2乘客疑染病



2018-04-05

〔記者林惠琴、方志賢 / 綜合報導〕麻疹群聚感染擴大！

日前衛福部疾管署公布今年國內首例境外移入麻疹，新北市三十餘歲男性在泰國感染、日本沖繩確診，返國後，除傳染給同事、搭機飛往日本傳染給台北市三十多歲的台灣虎航男空服員外，昨再確認當時同班機、台北市二十多歲女空服員也中鏢，同班機還有兩名乘客出現疑似症狀檢驗中。這起群聚事件追蹤接觸者已擴及一三五七人。



疾管署副署長莊人祥表示，該名女空服員幼時曾接種麻疹、腮腺炎、德國麻疹混合疫苗（MMR），一開始症狀不典型，但四月一日自覺發冷不適，即在家休息、未執行勤務，二日出現發燒、出疹症狀就醫，昨已確診麻疹，目前居家隔離中。

注目新聞

- 1 10月颱「哈吉貝」將生成？強度恐超越今年所有颱風
- 2 馬俊麟淚崩認愛王瞳 民視不忍了！開釗滾出八點檔
- 3 「香港人，反抗」！上千港人震撼宣讀《香港臨時政府宣言》
- 4 《禁蒙面法》今生效宛如戒嚴！香港全城失控、港鐵停駛
- 5 美移民局：國籍可寫「台灣」不會顯示「中華民國台灣」



Outline

- Introduction to Occupational Infectious diseases
- Occupational health hazards in health care workers
 - ✓ Needle-stick injury(針扎) – hepatitis B virus 、 hepatitis C virus 、 HIV/AIDS
 - ✓ Respiratory infection – tuberculosis
 - ✓ SARS, SARS-CoV-2
 - ✓ Ebola
- Standard precautions in health care

List of Occupational Diseases, ILO (revised 2010)

- Biological agents and infectious or parasitic diseases
 - 1. Brucellosis (布氏桿菌病,人畜共通, 牛、豬、山羊等動物)
 - 2. Hepatitis viruses 肝炎
 - 3. Human immunodeficiency virus (HIV)
 - 4. Tetanus (破傷風)
 - 5. Tuberculosis 肺結核
 - 6. Toxic or inflammatory syndromes associated with bacterial or fungal contaminants
 - 7. Anthrax (炭疽病)
 - 8. Leptospirosis (鉤端螺旋體病,人畜共通, 哺乳類動物)
 - 9. Diseases caused by other biological agents at work not mentioned in the preceding items where a direct link is established scientifically, or determined by methods appropriate to national conditions and practice, between the exposure to these biological agents arising from work activities and the disease(s) contracted by the worker

勞工保險職業病種類表

The List of Occupational Diseases for Labor Insurance (勞工保險條例第三十四條附表)

	職業病名稱	適用職業範圍
1	外爾氏病 (Weil's disease) 。 (severe form of leptospirosis)	有感染外爾氏病之工作場所。 (屠宰場、畜牧業、酪農業)
2	恙蟲病。	戶外勞動易患恙蟲病之工作場所。
3	豬型丹毒、炭疽、鼻疽等疾病。	接觸患病之動物、動物屍體、獸毛、生皮革及其他動物性之製品之工作場所。
4	從事醫療業務，由患者之病原體因接觸而引起之法定傳染病以外之傳染性疾病。	診療、治療及看護因職務之原因必須接觸患者之工作場所。

增列勞工保險職業病種類項目

	職業病名稱	適用職業範圍、工作場所或作業
2.1	退伍軍人症 (Legionellosis)	從事冷卻水塔維修、牙科門診等工作或工作於中央空調辦公室、旅館、醫院、安養院、精神病院、漩渦水療等有感染退伍軍人症之虞的工作場所。
2.2	漢他病毒出血熱	從事經常接觸嚙齒類動物之工作或工作於嚙齒類動物出沒頻繁等有感染漢他病毒出血熱之工作。
2.3	病毒性肝炎	醫療保健服務業工作人員因針扎、噴濺等途徑，或其他因工作暴露人體血液、體液導致感染之後所致。
2.4	結核病	從事必須接觸結核病患者或其檢體或廢棄物之工作。
2.5	後天免疫缺乏症候群	從事必須接觸愛滋病患者或其檢體或廢棄物之工作。
2.6	嚴重急性呼吸道症候群 (SARS)	從事必須接觸嚴重急性呼吸道症候群(SARS)患者或其檢體或廢棄物之工作。
2.7	Q型熱(Q fever)	在牛羊畜牧養殖業、屠宰場、相關實驗室、羊毛處理廠等作業環境工作，因而接觸到動物、動物屍體、或其未經消毒的產品。
2.8	登革熱	限於因職務性質所需，在蚊蟲聚集的草叢水渠等地『例行、經常性、規律地』工作之人員。
2.9	禽流感	在家禽養殖場、屠宰場、相關實驗室等作業環境工作，因而遭受病禽所散播的病毒感染。

Work-Relatedness of Health and Disease

WHO Definition (1985)

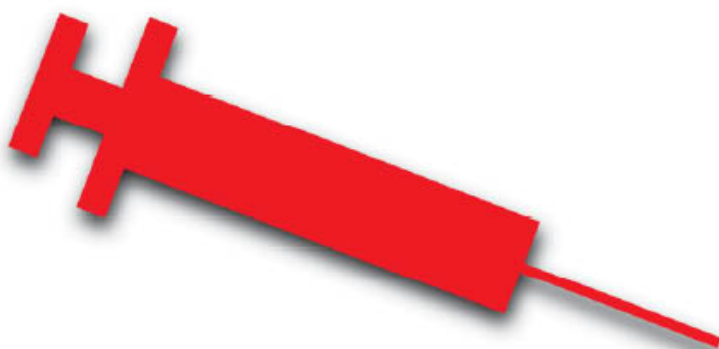
- 職業是疾病之必要原因
 - Causative factors of occupational diseases
 - e.g. Pneumoconiosis, mesothelioma
- 職業是疾病之影響因素
 - *Multi-factorial work-related diseases*
 - 職業會引發或惡化潛伏的身心狀態
 - e.g. Asthma, cardiovascular disease
- 工作環境中有潛在的危險物質
 - General diseases affecting working population
 - e.g. Influenza

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- Introduction to Occupational Infectious diseases
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- Respiratory infection - tuberculosis
- Skin infection – Scabies
- SARS
- Ebola
- Standard precautions in health care

針扎 (Needle stick injuries)

Sharpsafe®



SHARPS INJURY

DEALING WITH NEEDLESTICK INJURIES

PREVENT NEEDLESTICK INJURY	WHAT TO DO
① Never resheath a needle.	① Squeeze the wound and make it bleed
② Always use Near Patient Sharps System.	② Clean the wound thoroughly with running water
③ Always dispose of sharps at point of use.	③ Apply a waterproof dressing
④ Always close sharps containers when 2/3 full.	④ Inform your line manager
⑤ Ensure source and date are completed on the sharps box.	⑤ Complete Incident Report Forms
	⑥ Report to Occupational Health or A&E
	⑦ Refer to Policy for more details

IS YOUR HEPATITIS B VACCINATION UP TO DATE?

For information on needlestick injuries please contact:

frontiermedical® group

www.frontiermedical.eu



表 7 2011~2018 年納入醫院歷年通報扎傷數(N=7,130)

醫院類型	2011	2012	2013	2014	2015	2016	2017	2018	總計
醫院層級									
醫學中心	639	628	560	593	595	538	450	504	4,507
區域醫院	306	308	238	300	302	272	257	240	2,223
地區醫院	45	56	51	60	47	40	50	51	400
公私立別									
公立	544	491	428	505	517	450	443	452	3,830
私立	446	501	421	448	427	400	314	343	3,300
教學醫院									
是	979	977	835	943	930	842	747	781	7,034
否	11	15	14	10	14	8	10	14	96
總計	990	992	849	953	944	850	757	795	7,130

針扎職業危害預防策略成效評估研究

- 109 年度研究計畫 ILOSH108-A303

表 8 2011~2018 年各職類工作人員歷年扎傷事件分布情形(N=7,128)

職類別	2011	2012	2013	2014	2015	2016	2017	2018	總計
通報案件數(百分比)									
醫師	267 (27.0)	249 (25.1)	207 (24.4)	245 (25.7)	279 (29.6)	263 (31.0)	220 (29.1)	226 (28.4)	1,956
護理人員	585 (59.0)	597 (60.2)	508 (59.8)	550 (57.7)	515 (54.5)	468 (55.1)	428 (56.6)	450 (56.6)	4,101
醫事技術人員	53 (5.4)	55 (5.5)	67 (7.9)	81 (8.5)	68 (7.2)	63 (7.4)	65 (8.6)	55 (6.9)	507
支援人員	85 (8.6)	91 (9.2)	67 (7.9)	77 (8.1)	82 (8.7)	55 (6.5)	43 (5.7)	64 (8.1)	564
總計	990 (100.0)	992 (100.0)	849 (100.0)	953 (100.0)	944 (100.0)	849 (100.0)	756 (100.0)	795 (100.0)	7,128

遺漏值：2 件未填寫工作人員職業別

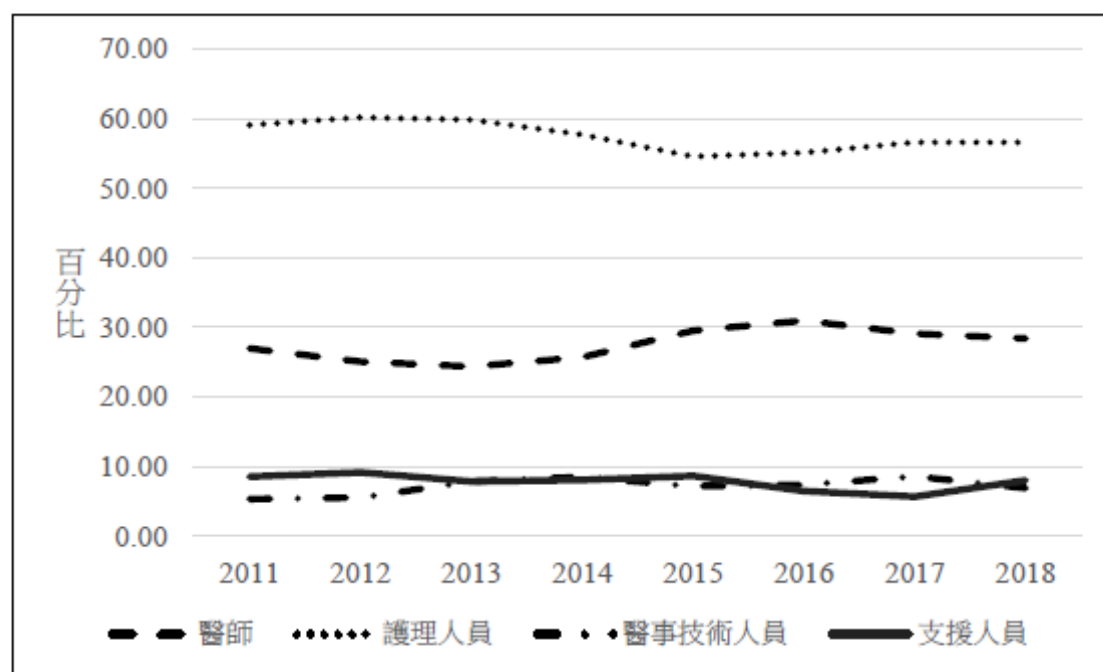


圖 3 2011~2018 年各職類工作人員歷年扎傷事件分布趨勢圖(N=7,128)

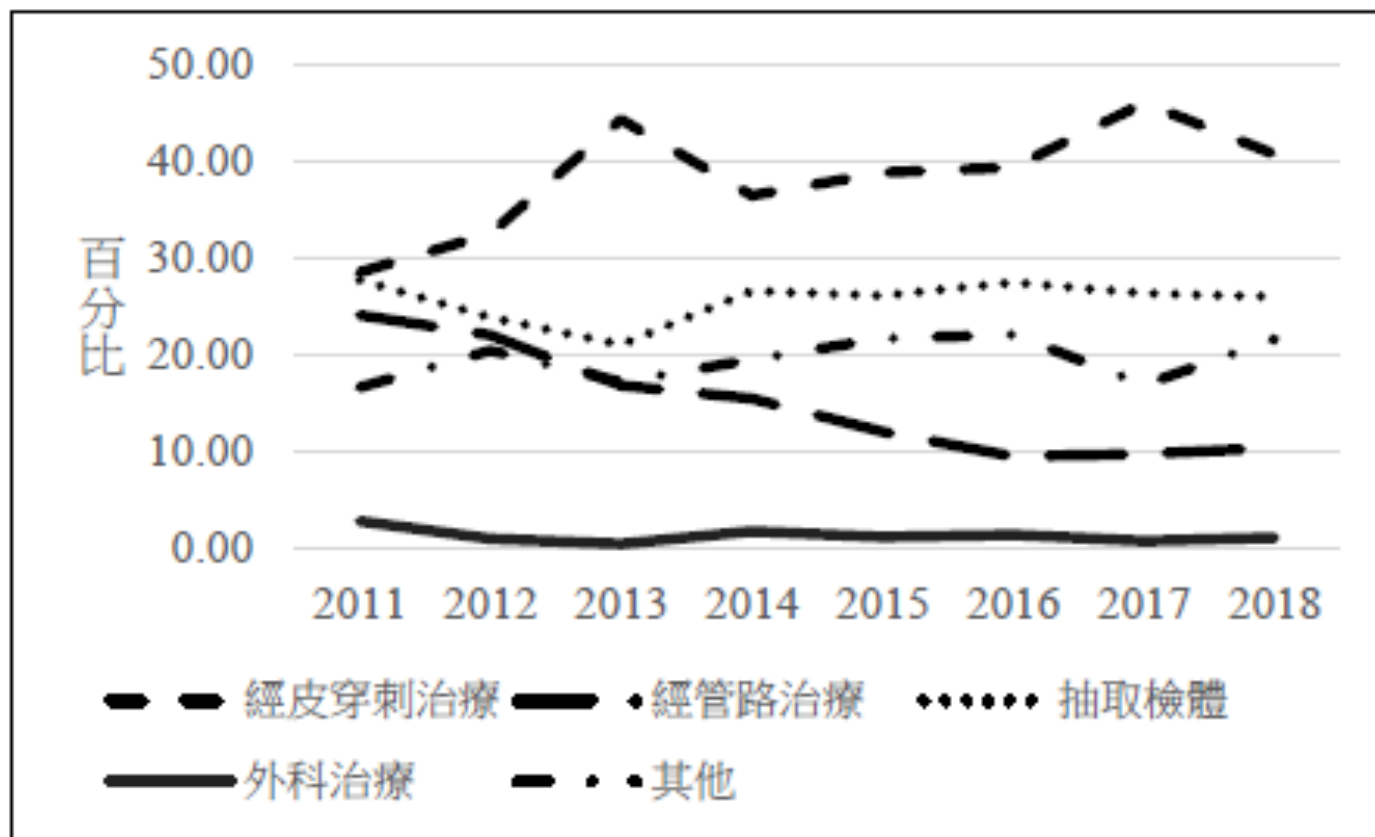


圖 4 2011~2018 年針頭或銳物原始用途扎傷事件分布趨勢圖(N=7,130)

針扎職業危害預防策略成效評估研究

- 109 年度研究計畫 ILOSH108-A303

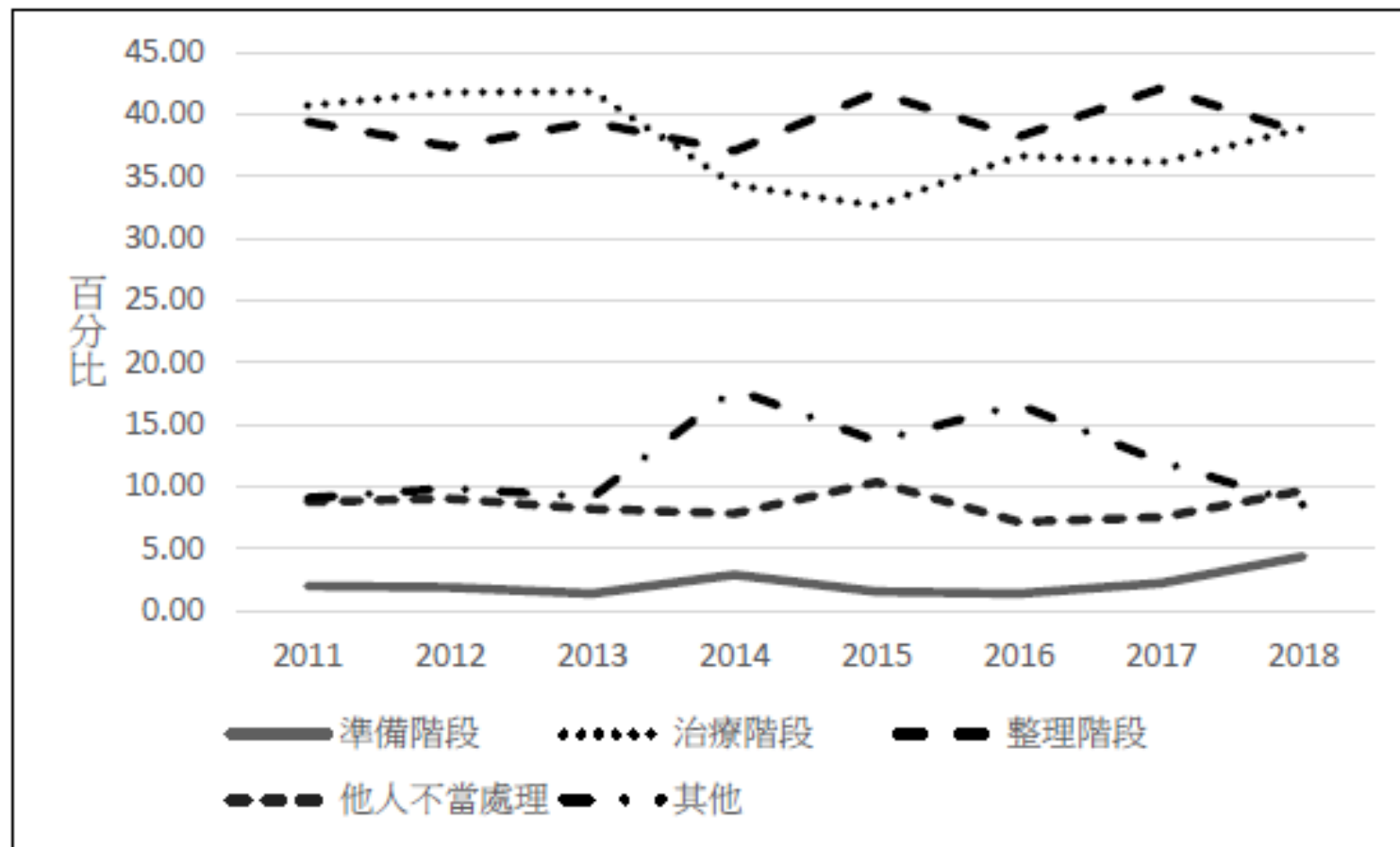


圖 5 2011~2018 年扎傷事件發生情境分布趨勢圖(N=7,130)

針扎職業危害預防策略成效評估研究

- 109 年度研究計畫 ILOSH108-A303

表 11 2011~2018 年引起扎傷事件之針具安全設計分布情形(N=7,130)

針具	2011	2012	2013	2014	2015	2016	2017	2018	總計
通報案件數(百分比)									
一般針具	785 (79.3)	794 (80.0)	711 (83.8)	794 (83.3)	755 (80)	659 (77.5)	567 (74.9)	456 (57.4)	5,521
安全針具	48 (4.8)	85 (8.6)	75 (8.8)	131 (13.8)	124 (13.1)	126 (14.8)	147 (19.4)	186 (23.4)	922
不知道	157 (15.9)	113 (11.4)	63 (7.4)	28 (2.9)	65 (6.9)	65 (7.7)	43 (5.7)	153 (19.2)	687
總計	990 (100.0)	992 (100.0)	849 (100.0)	953 (100.0)	944 (100.0)	850 (100.0)	757 (100.0)	795 (100.0)	7,130

針扎職業危害預防策略成效評估研究

- 109 年度研究計畫 ILOSH108-A303

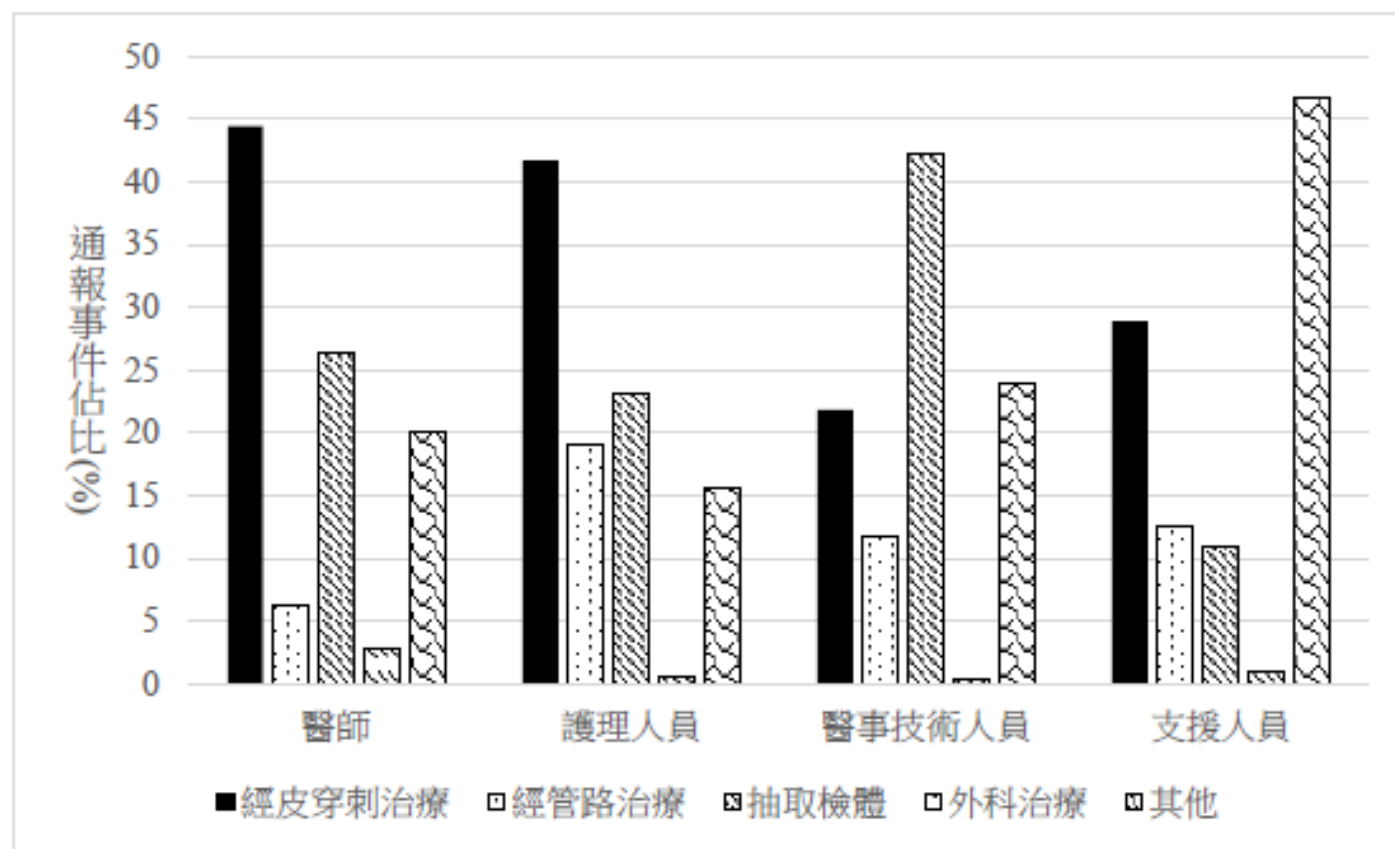
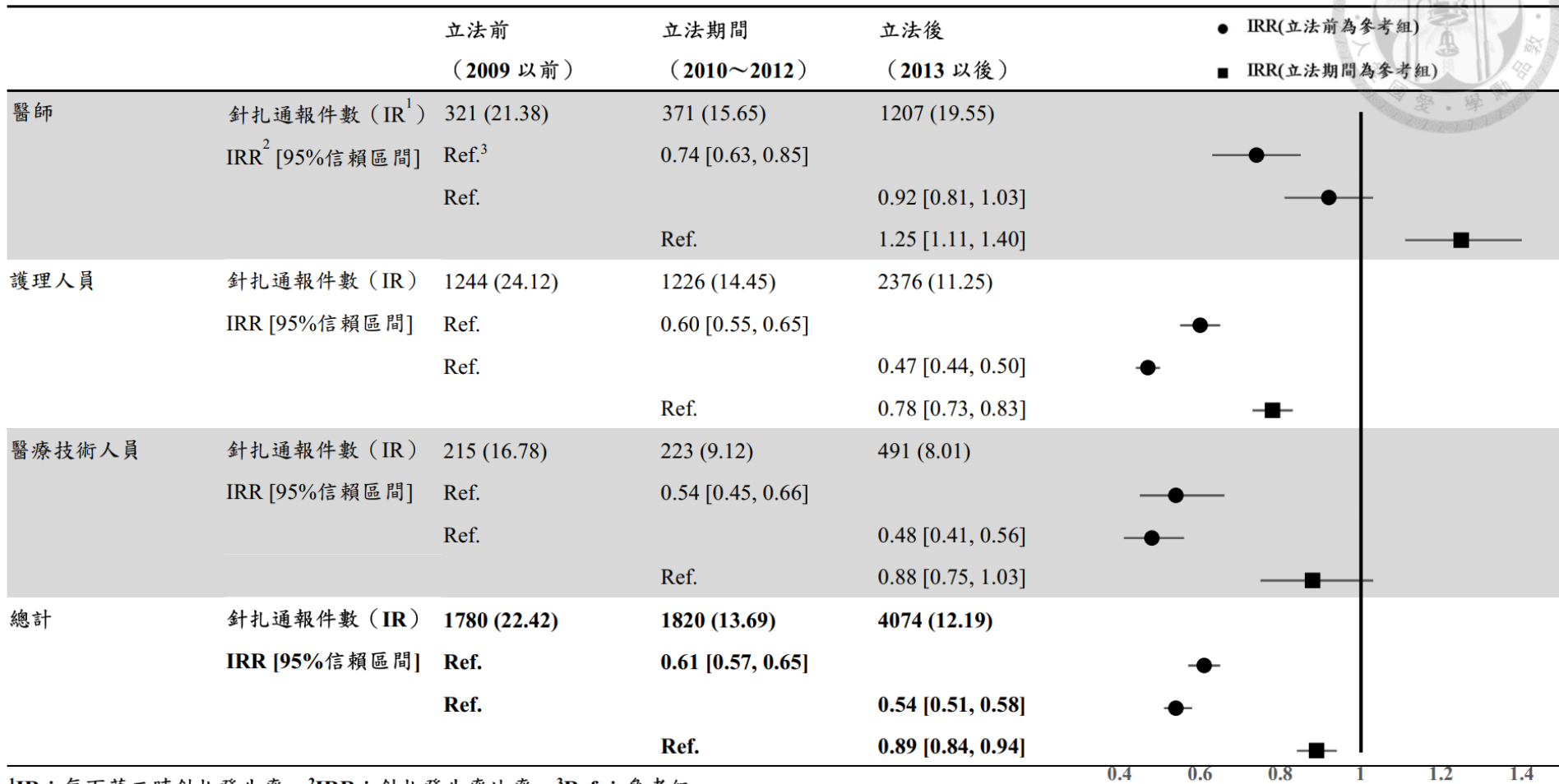


圖 8 2011~2018 年一般針具扎傷事件原始用途分布圖(N=5,521)

表 9 各職類立法前後針扎發生率列表 (N=7,674)



¹IR：每百萬工時針扎發生率 ²IRR：針扎發生率比率 ³Ref.：參考組

2011年12月修正醫療法規定，「醫療機構對於所屬醫事人員執行直接接觸病人體液或血液之醫療處置時，應自中華民國101年起，五年內按比例逐步完成全面提供安全針具」

WORK PRACTICES WHICH INCREASE THE RISK OF NEEDLE STICK INJURY

- Recapping needles (Most important)
- Performing activities involving needles and sharps in a hurry
- Handling and passing needles or sharp after use
- Failing to dispose of used needles properly in puncture resistant sharps containers
- Poor healthcare waste management practices
- Ignoring Universal Work Precautions

Hepatitis B in Taiwan



- Hepatitis B virus carriers BH
- Prevalence in Taiwan before 1984: 15-20% in adults
- Transmission route in Taiwan
 - Vertical transmission from mothers: 60%
 - Horizontal transmission: 40%
 - Sexual contact
 - Injecting drug use or other percutaneous exposure i.e. tattoos
- Vaccination
 - 1986: Universal vaccination program for infants

Acute infection

- Incubation period: 1 – 4 months
- Majority: subclinical or anicteric hepatitis (need blood test to confirm)
- Symptomatic infection: 30-50% in adults, less than 10% in children;
- 症狀 Fatigue, anorexia, nausea, jaundice, and right upper quadrant discomfort
- 0.1-0.5%: Fulminant hepatic failure
- There are two possible outcomes following hepatitis B infection
 - Clearance of the hepatitis B virus
 - Chronic hepatitis B infection

Fulminant hepatitis

- Very high mortality
- Before Taiwan implemented universal hepatitis B vaccination, every cohort of medical interns or newly recruited nurses had cases of needlestick injuries leading to fulminant hepatitis.



Chronic hepatitis 慢性肝炎

- Risk to develop chronic hepatitis
 - Newborn: 90%;
 - 1-5 y/o children: 20-50%
 - >6 y/o: 1-10%
- Prognosis
 - Hepatocellular carcinoma: RR=14-20
 - Estimated five-year rates of progression
 - Chronic hepatitis to cirrhosis : 10 – 20% 肝硬化
 - Compensated cirrhosis to hepatic decompensation: 20 – 23%
 - Compensated cirrhosis to HCC : 6 – 15%



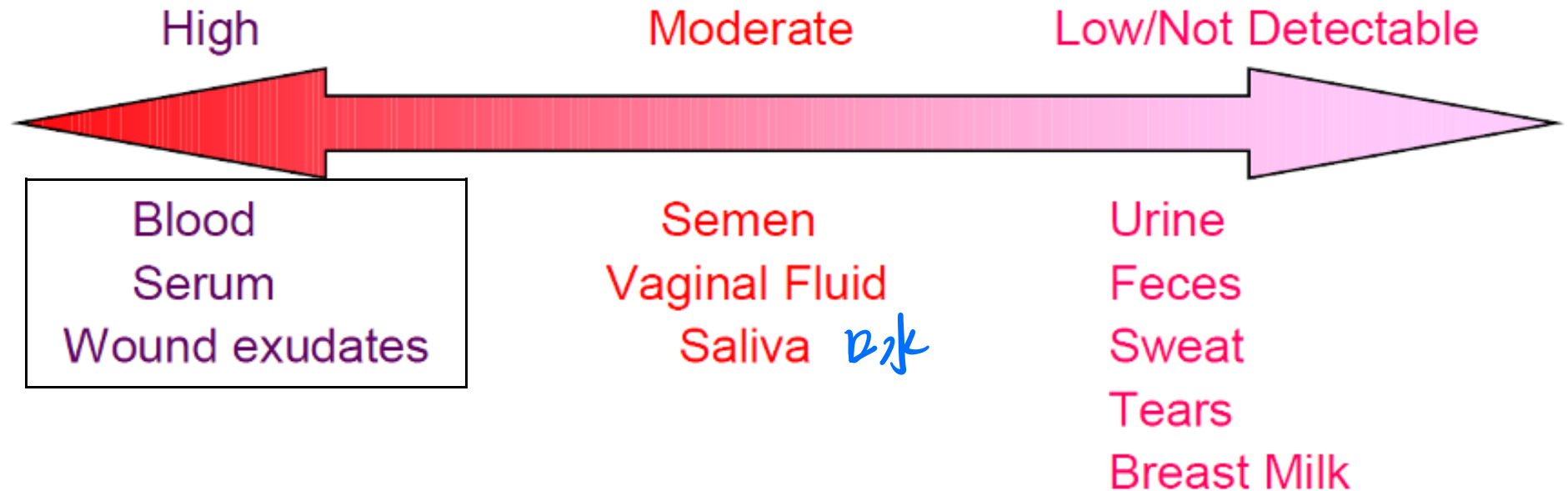
HBV serologic markers (1)

- Hepatitis B surface antigen (表面抗原, HBsAg) → 現在有B肝
- The presence of HBsAg indicates that the person is infectious, carrier
- Chronic carriers: HBsAg persists for more than 6 months
- Hepatitis B surface antibody (表面抗體, anti-HBs, ~~HBsAb~~) → 有免疫力
- Indicates the person has recovered from and has immunity to Hepatitis B or has been vaccinated.
- Have immunity against hepatitis virus
- Hepatitis B core antibody (核心抗體, anti-HBc)
- previous or ongoing infection

HBV serologic markers (2)

- Hepatitis B envelope antigen (e抗原, HBeAg)
 - Acute and chronic Hepatitis B infection.
 - Indicates that the virus is replicating and the infected person has high levels of the Hepatitis B virus
 - Highly infectious
- Hepatitis B envelope antibody (e抗體, anti-Hbe)
 - Indicates lower levels of HBV.
 - Spontaneous conversion from e antigen to e antibody (a change known as “e” seroconversion) is a predictor of longterm clearance of HBV in patients undergoing antiviral therapy
 - Spontaneous “e” seroconversion also occurs in natural infection

HBV Concentration in human



Hepatitis B control in Taiwan

- 1984-1986: infants of mothers with positive HBsAg
 - vaccination and 0.5ml of hepatitis B immunoglobulin
- 1986: the world's first nationwide hepatitis B virus (HBV) universal vaccination
- Seroprevalence of HBsAg in children declined from 9.8% to 0.6% after 20 years of mass vaccination
- Incidence of HCC in 6- to 9-year-olds declined from 0.52 per 100,000 for those born between 1974 and 1984 to 0.13 for those born between 1984 and 1986 ($p < 0.001$)

台灣醫療人員B型肝炎疫苗接種政策

- 無B型肝炎抗原或抗體之新進及在職醫療人員均必須接受三劑B型肝炎疫苗注射，並接受抗體追蹤及後續處置
- 無B型肝炎抗原或抗體之員工若拒絕接種B型肝炎疫苗，應簽署切結書，日後若出現高風險針扎之狀況，醫院不予給付免疫球蛋白費用

(**醫療機構執行感染控制查核辦法**：醫療機構應訂有員工保健計畫，提供預防接種、體溫監測及胸部 X 光等 必要之檢查或防疫措施；並視疫病防治需要，瞭解員工健康狀況，配合提供必要措施。醫療機構應訂定員工暴露病人血液、體液及尖銳物品扎傷事件之預防、追蹤及處置標準作業程序。)

Hepatitis C virus

C肝

- **Acute infection**
 - 20-30% symptomatic
 - Incubation period: 2 weeks to 6 months
 - Fever, fatigue, anorexia, nausea, jaundice, abdominal pain
 - Fulminant hepatitis 暴发型肝炎
- **Chronic infection**
 - 60%-80% develop chronic hepatitis.
 - About 5-20% of people with chronic infection develop liver cirrhosis after 20-30 years;
 - About 1-5% people die due to liver cirrhosis or liver



為什麼要治療C肝

沒疫苗

1

C肝是可以治癒的

B肝雖有疫苗可以預防，但目前的藥物治療只能控制，無法根治。而目前的C肝全口服新藥，治癒率高達9成以上，甚至逼近百分之百。



2

避免走向肝硬化、肝癌

及早開始治療C肝，可避免肝臟細胞一再發炎而產生纖維化，甚至走向肝硬化，也可降低肝癌發生的機率。



3

治療就是預防

C肝並無疫苗可以預防，治癒C肝，可避免將C肝病毒傳染給他人，就能阻斷C肝病毒的傳播。



4

減少醫療資源的耗費

C肝所衍生的肝硬化、肝癌，會增加很多醫療資源的耗費，當治癒C肝、減少傳染源，就可省下更龐大的醫療支出。



三大核心策略及三大政策方向

精準公衛

防治一條龍

防治在地化

以治療引領預防

以篩檢支持治療

以預防鞏固成效

2025
消除C肝

↑ C型肝炎治療涵蓋率

↓ C型肝炎病毒新感染

↓ C型肝炎病毒相關死亡

WHO 消除 C 肝之各項目標與台灣現況

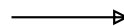
計畫性目標	WHO 2030年 認證目標	台灣 2025年 現況	WHO制訂之 C型肝炎的公共 衛生消除標準 (PTE)*
C肝 (HCV RNA) 診斷率	≥ 90%	90.5% 45歲~84歲	金 (≥ 80%)
被診斷C肝者治療率	≥ 80%	92.5%	金 (≥ 70%)
C肝預防	安全注射率 100%	已達到	金 (100%)
	血品安全 100%	已達到	金 (100%)
	每位靜脈注射藥癮者 可取得的安全針具300份	目前為 292份 安全針 具 (截至2023)	金 (≥ 150份)
影響性目標			
B、C肝相關之肝癌及失代償 性肝硬化造成的死亡率	≤ 6/10萬人	尚未達到	* 係在2030年前鼓 勵各國持續努力的 階段性認證措施
新感染C肝年發生率 (一般族群)	≤ 5/10萬人	2.85 /10萬人	
新感染C肝年發生率 (靜脈注射藥癮族群)	≤ 2%	待確認	

Natural History of HIV infection

Virus exposure



**Acute infection
(2-4 weeks)**



Fever, rash, chills, sore throat, fatigue, loss of appetite, swollen lymph nodes, muscle pain. Only 50% are symptomatic.



**Incubation period
(5-15 years)**



Few or no symptoms.
Slow viral-mediated CD4+ T-cell destruction



**AIDS
(2-5 years)**



CD4+ T cell count below 200 cells per μL .
Opportunistic infections: pneumocystis pneumonia, esophageal candidiasis

Risk of infection

與愛滋病毒感染者發生危險行為之危險性

Box 2.2 Risk of HIV transmission by exposure category

Type of exposure	Risk
<i>Occupational</i>	
Needle stick injury	0.3%
Mucous membranes contact	0.09%
<i>Non-occupational</i>	
Receptive anal intercourse	1-30%
Insertive anal intercourse	0.1-10%
Receptive vaginal intercourse	0.1-10%
Insertive vaginal intercourse	0.1-1%
Needle sharing	0.67%

Healthcare Personnel with Documented and Possible Occupationally Acquired HIV Infection, by Occupation, 1981-2013

TABLE. Number of confirmed or possible cases of occupationally acquired HIV infection among health care workers reported to CDC — United States, 1985–2013

Occupation	Confirmed (N = 58)		Possible (N = 150)	
	No.	(%)	No.	(%)
Nurse	24	(41.4)	37	(24.7)
Laboratory technician, clinical	16	(27.6)	21	(14.0)
Physician, nonsurgical	6	(10.3)	13	(8.7)
Laboratory technician, nonclinical	4	(6.9)	—	—
Housekeeper/maintenance	2	(3.4)	14	(9.3)
Technician, surgical	2	(3.4)	2	(1.3)
Embalmer/morgue technician	1	(1.7)	2	(1.3)
Hospice caregiver/attendant	1	(1.7)	16	(10.7)
Respiratory therapist	1	(1.7)	2	(1.3)
Technician, dialysis	1	(1.7)	3	(2.0)
Dental worker, including dentist	—	—	6	(4.0)
Emergency medical technician/paramedic	—	—	13	(8.7)
Physician, surgical	—	—	6	(4.0)
Technician/Therapist, other	—	—	9	(6.0)
Other health care occupations	—	—	6	(4.0)

Abbreviation: HIV = human immunodeficiency virus.

Source: Division of HIV/AIDS Prevention, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention, CDC.

- **Possible case:**
- have no other risk factors for HIV infection
- have a history of occupational exposure to blood, body fluids, or HIV-infected laboratory material
- Seroconversion after exposure was not documented

表 36 護理人員汙染性針扎之追蹤檢驗結果

檢驗項目	陽轉 n(%)	陰性 n(%)	已有抗體	原本即為帶原者	遺漏值
B 型肝炎	4 (10.8)	33 (89.2)	217	19	138

檢驗項目	陽轉 n(%)	陰性 n(%)	原本即為帶原者	遺漏值
C 型肝炎	4 (1.6)	241 (98.4)	1	165
HIV	1 (0.4)	244 (99.6)	0	166
梅毒	0 (0.0)	245 (100)	1	165

安全針具立法實施後針扎職業危害探討研究

- 107 年度研究計畫 ILOSH 107 - A308

Survival of HIV-positive patients starting antiretroviral therapy between 1996 and 2013: a collaborative analysis of cohort studies

The Antiretroviral Therapy Cohort Collaboration*



Summary

Background Health care for people living with HIV has improved substantially in the past two decades. Robust estimates of how these improvements have affected prognosis and life expectancy are of utmost importance to patients, clinicians, and health-care planners. We examined changes in 3 year survival and life expectancy of patients starting combination antiretroviral therapy (ART) between 1996 and 2013.

Methods We analysed data from 18 European and North American HIV-1 cohorts. Patients (aged ≥ 16 years) were eligible for this analysis if they had started ART with three or more drugs between 1996 and 2010 and had at least 3 years of potential follow-up. We estimated adjusted (for age, sex, AIDS, risk group, CD4 cell count, and HIV-1 RNA at start of ART) all-cause and cause-specific mortality hazard ratios (HRs) for the first year after ART initiation and the second and third years after ART initiation in four calendar periods (1996–99, 2000–03 [comparator], 2004–07, 2008–10). We estimated life expectancy by calendar period of initiation of ART.

Findings 88 504 patients were included in our analyses, of whom 2106 died during the first year of ART and 2302 died during the second or third year of ART. Patients starting ART in 2008–10 had lower all-cause mortality in the first year after ART initiation than did patients starting ART in 2000–03 (adjusted HR 0·71, 95% CI 0·61–0·83). All-cause mortality in the second and third years after initiation of ART was also lower in patients who started ART in 2008–10 than in those who started in 2000–03 (0·57, 0·49–0·67); this decrease was not fully explained by viral load and CD4 cell count at 1 year. Rates of non-AIDS deaths were lower in patients who started ART in 2008–10 (vs 2000–03) in the first year (0·48, 0·34–0·67) and second and third years (0·29, 0·21–0·40) after initiation of ART. Between 1996 and 2010, life expectancy in 20-year-old patients starting ART increased by about 9 years in women and 10 years in men.

Interpretation Even in the late ART era, survival during the first 3 years of ART continues to improve, which probably reflects transition to less toxic antiretroviral drugs, improved adherence, prophylactic measures, and management of comorbidity. Prognostic models and life expectancy estimates should be updated to account for these improvements.

Funding UK Medical Research Council, UK Department for International Development, EU EDCTP2 programme.

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See [Comment](#) page e324

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最新研究顯示，一名20歲的愛滋病病毒感染者在2008年或以後開始進行抗逆轉錄病毒治療，一年治療後病毒載量低，預期壽命接近一般人口，約78歲。

Risk of transmission through needle stick injury

血液病源病毒經針扎之傳染力

Source	Risk
HBV(HBsAg+)	
HBeAg+	22.0%-31.0% clinical hepatitis 37.0%-62.0% serological evidence of HBV infection
HBeAg-	1.0%- 6.0% clinical hepatitis; 23.0%-37.0% serological evidence of HBV infection
HCV	1.8% (0%-7%)
HIV	0.3% (0.2%-0.5%) [Lauer et al., 2001]

被針頭或銳物扎傷時

1. 立刻將傷口周圍血液擠出
2. 以大量清水沖洗傷口
3. 告並填寫EPINet之「針頭和銳物扎傷事件報告單」及「追蹤報告單」並領取職業傷病門診就診單。

皮膚或黏膜暴露於病患之血液、體液*時

(*如精液、陰道分泌物、腦脊髓液、心包液、肋膜液、腹膜液、關節液、及羊水等)

1. 以流動清水或0.9%生理食鹽水(若有傷口)沖洗
2. 告並填寫EPINet之「血液、體液暴露事件報告單」及「追蹤報告單」並領取職業傷病門診就診單。

檢查醫療人員和病患之抗原及抗體

1. 針扎來源：HbsAg、anti-HCV、anti-HIV (PA)、Syphilis (VDRL、TPHA)
2. 被針扎者：HbsAg、anti-HBs、anti-HCV、anti-HIV (PA)、Syphilis (VDRL、TPHA)、GOT、GPT

病患 HIV (+)

醫療人員應儘速檢驗Anti-HIV，並由醫師判定是否於36hrs. 內投予預防性用藥

暴露後6週、3、6個月定期追蹤Anti-HIV

註：病患HIV與Anti-HCV同時呈現(+)則延長追蹤至一年

病患 HbsAg (+) 或 Anti-HBc (+)

醫療人員
HbsAg (+) 或
Anti-HBs (+)

不需注射疫苗或
HBIG

醫療人員
HbsAg (-)
Anti-HBs (-)
未注射疫苗

24小時內注射HBIG

醫療人員
HbsAg (-)
Anti-HBs (-)
已完成疫苗注射
未產生抗體

24小時內注射HBIG
並補一劑疫苗

醫療人員
HbsAg (-)
Anti-HBs (-)
正接受疫苗注射
未產生抗體

注射HBIG並繼續完成疫苗注射

暴露後6個月、一年，
回門診追蹤HbsAg、Anti-HBs、Anti-HBc、GOT、GPT

病患 Anti-HCV (+)

醫療人員
Anti-HCV (+)

暴露後3、6個月，
繼續追蹤肝功能
GOT、GPT

醫療人員
Anti-HCV (-)

暴露後3、6個月，
一年定期追蹤GOT、
GPT、Anti-HCV

病患 VDRL (+) TPHA ≥ 1:160

建議醫療人員至感
染科追蹤治療

一年內後如無症狀
發生者，可停追蹤

* 污染源不明，視同陽性處理。



Work Practice Controls

- Contaminated needles/sharps shall not be bent, ➡ 不要回套
recapped or removed unless there is no feasible
alternative or if required by a specific medical
procedure
- Such bending, recapping, or removal must be done
through use of mechanical device or a one
technique
- Use puncture-resistant sharps
container for disposal of sharps





台灣早期的針扎政策

- 以醫療人員B型肝炎疫苗接種為主軸
- 強調針扎後之傷害救濟
- 缺點
 - 針扎導致的疾病不是只有B型肝炎
 - 不論疫苗接種或是傷害救濟都沒在本質上做出減少針扎的努力

台灣中期的針扎政策

- 以改善醫院的管理為主軸
- 以針扎通報來調查針扎發生的原因，並執行改善作為
- 均已列入醫院評鑑的項目中
- 缺點
 - 執行到位不易，針扎事件仍經常發生
 - 針扎原因之探討仍為表象，對實質減少針扎幫忙不大，同時也無法預防意外之發生

台灣後期的針扎政策

- 以推動安全針具為主軸
- 先由針扎好發區域進行突破，以試辦成效說服領導進行全面使用
- 立法

醫療法修正

- 醫療法第56條及第101條條文修正對照表

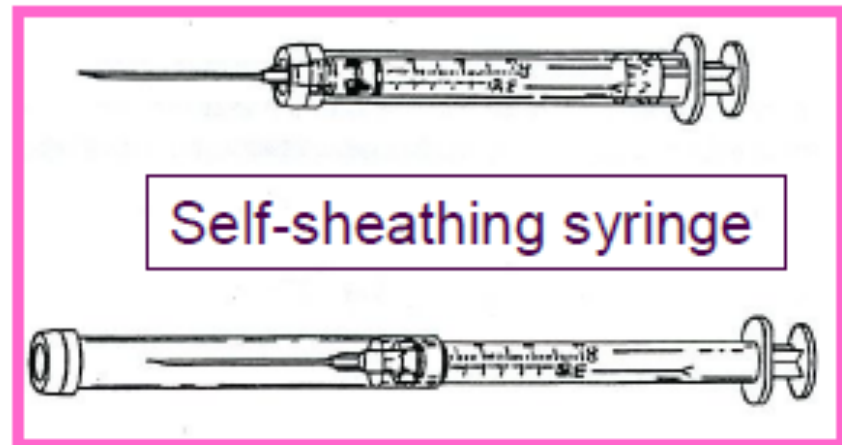
條號	原條文(2009年5月20日)	修正後(2011年12月21日)
第56條	醫療機構應依其提供服務之性質，具備適當之醫療場所及安全設施。	醫療機構應依其提供服務之性質，具備適當之醫療場所及安全設施。 醫療機構對於所屬醫事人員執行直接接觸病人體液或血液之醫療處置時，應自中華民國一百零一年起，五年內按比例逐步完成全面提供安全針具。
第101條	違反第 <u>十七</u> 條第一項、第 <u>十九</u> 條第一項、第 <u>二十</u> 條、第 <u>二十二</u> 條第一項、第 <u>二十三</u> 條第一項、第 <u>二十四</u> 條第一項規定者，經予警告處分，並限期改善；屆期未改善者處新臺幣一萬元以上五萬元以下罰鍰。	違反第 <u>十七</u> 條第一項、第 <u>十九</u> 條第一項、第 <u>二十</u> 條、第 <u>二十二</u> 條第一項、第 <u>二十三</u> 條第一項、第 <u>二十四</u> 條第一項、第 <u>五十六</u> 條第二項規定者，經予警告處分，並限期改善；屆期未改善者，處新臺幣一萬元以上五萬元以下罰鍰， 按次連續處罰 。

安全針具

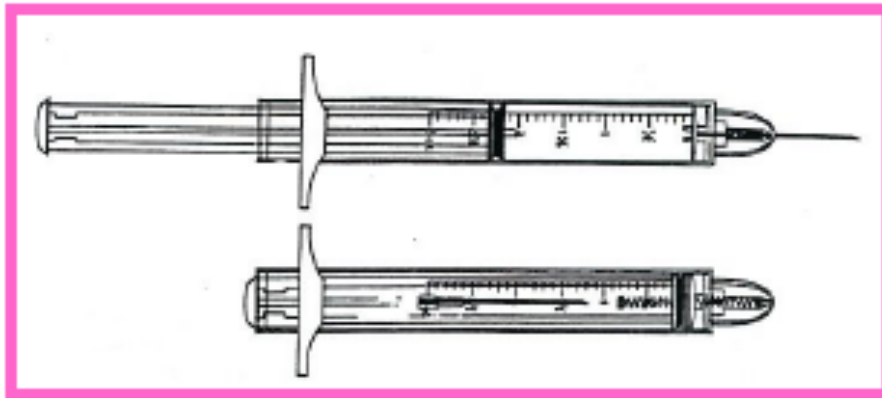
Engineering Controls for Needle Stick

- Sharps with Engineered Sharps Injury Protection (SESIP)—a non-needle sharp or needle with a built-in safety feature or mechanism that effectively reduces the risk of an exposure incident

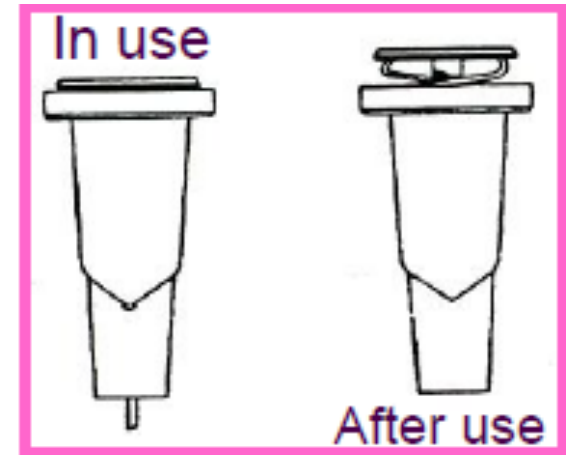
Examples include:



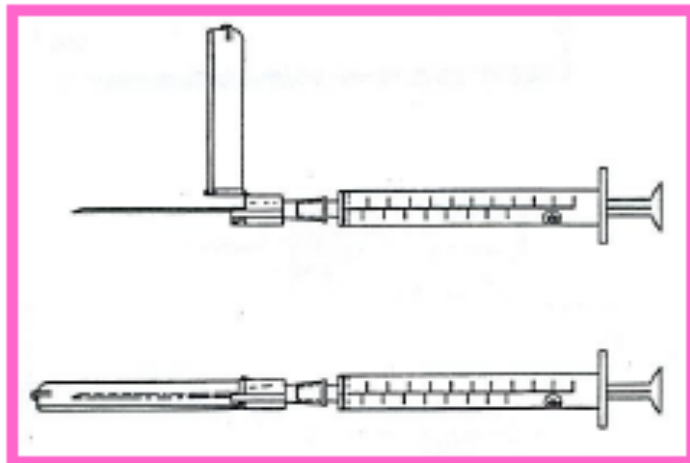
More Examples



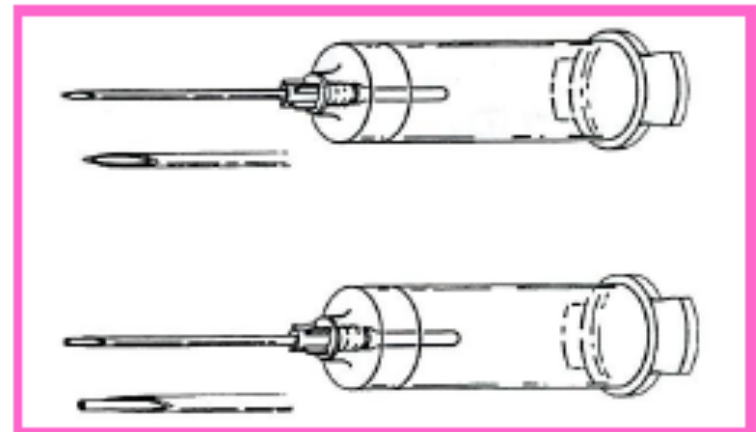
Retractable needle technology



Retractable lancets



Add-ons (needle covers)



Self-blunting needles

扎傷暴露 控制計畫

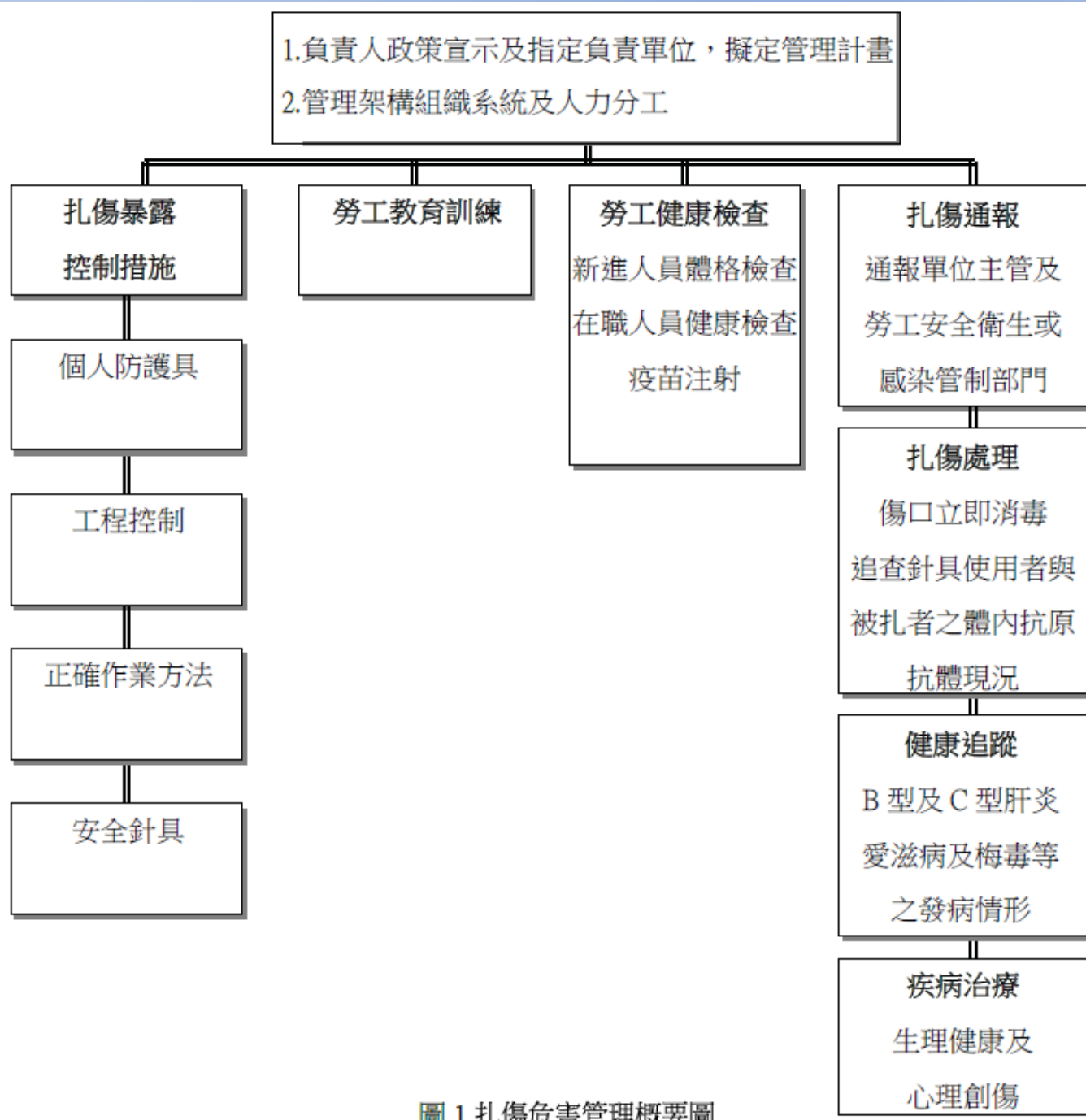


圖 1 扎傷危害管理概要圖

Outline

- Introduction to Occupational Infectious diseases
- Occupational health hazards in health care workers
- Needle-stick injury(針扎) – hepatitis B virus 、 hepatitis C virus 、 HIV/AIDS
- Respiratory infection - tuberculosis
- SARS, SARS-CoV-2
- Ebola
- Standard precautions in health care

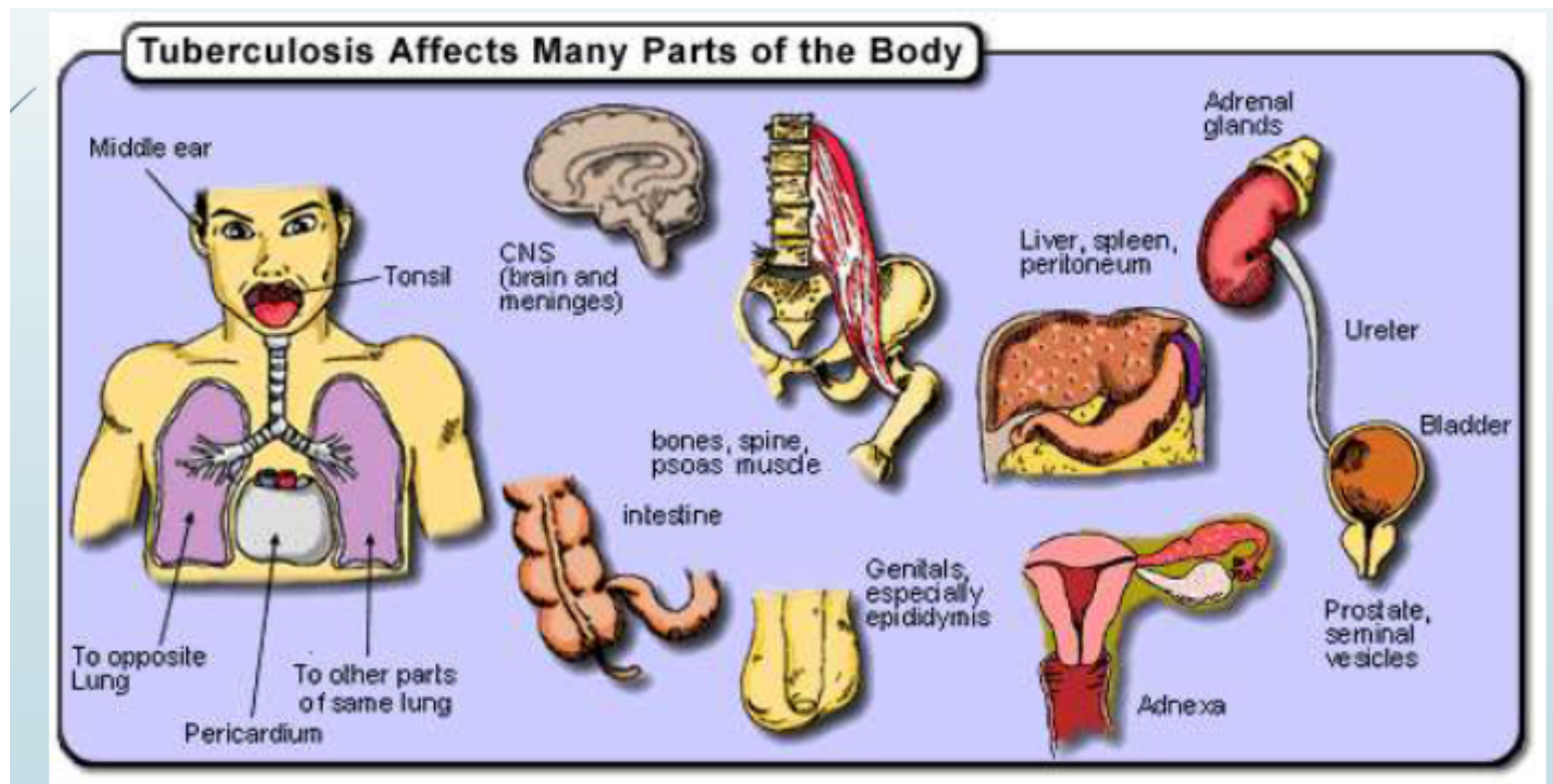
Tuberculosis 肺結核

Globally

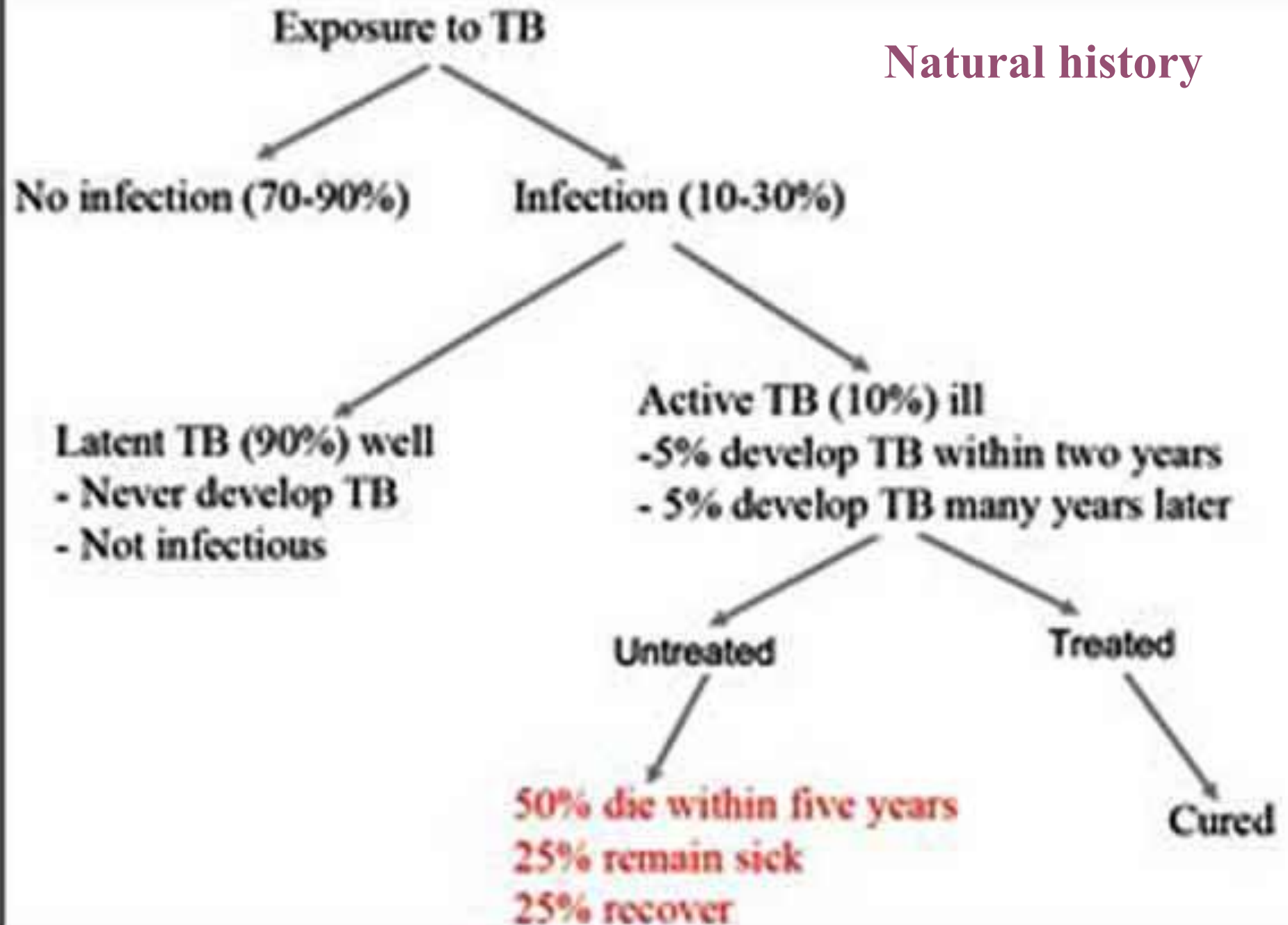
- Tuberculosis (TB) is one of the top 10 causes of death worldwide.
 - About one-third of the world's population has latent TB
 - In 2020, 5.8 million people fell ill with TB, and 1.51 million died from the disease (including 214000 among people with HIV). Over 95% of TB deaths occur in low- and middle-income countries.
 - Ending the TB epidemic by 2030 is among the health targets of the Sustainable Development Goals.
-
- In Taiwan
 - 6,584 new cases in 2023
 - 452 deaths in 2023

Tuberculosis

- A multi-systemic disease caused by bacteria (*Mycobacterium tuberculosis*)
- Pulmonary TB: 85%
- Extrapulmonary TB: 15%



Natural history

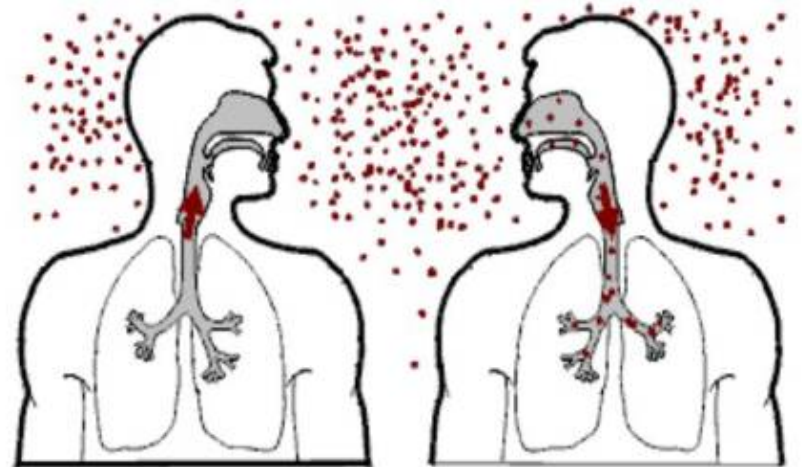


Symptoms of active TB

- Systematic : fatigue, loss of appetite, fever, chills, night sweats, weight loss,
- Pulmonary : cough with sputum and blood, chest pain
- 肺結核早期多半沒有明顯症狀，所以容易被忽略或延誤病情。
- 當發現身體不適，特別是有咳嗽兩週仍未痊癒、發燒、疲倦、夜間盜汗、食慾差等症狀者，有可能是感染結核菌，應儘早儘快就診，接受驗痰、胸部X光等檢查。

Transmission route

- ^{微粒}**Droplet nuclei** (1–5 microns in diameter)
- Tiny water droplets with the bacteria that are released when persons who have pulmonary or laryngeal tuberculosis cough, sneeze, laugh, shout etc.
- These tiny droplet nuclei remain suspended in the air for up to several hours.



Infectiousness

- **Low infectious dose** of less than 10 microorganisms
- A person with active TB cough can transmit the disease to 10 -15 people on average in a single year
- Factors associated with infectiousness of **TB patients**:

Factor	Description
Clinical	• Presence of cough, especially lasting 3 weeks or longer • Respiratory tract disease, especially with involvement of the larynx (highly infectious) • Failure to cover the mouth and nose when coughing • Inappropriate or inadequate treatment (drugs, duration)
Procedure	• Undergoing cough-inducing or aerosol-generating procedures (e.g., bronchoscopy, sputum induction, administration of aerosolized medications)
Radiographic and laboratory	• Cavitation on chest radiograph • Positive culture for <i>M. tuberculosis</i> • Positive AFB sputum smear result

Factors related to transmission: Environmental

Table 2.3
Environmental Factors that Enhance the Probability that
M. tuberculosis Will Be Transmitted

Factor	Description
Concentration of Infectious droplet nuclei	The more droplet nuclei in the air, the more probable that <i>M. tuberculosis</i> will be transmitted
Space	Exposure in small, enclosed spaces
Ventilation	Inadequate local or general ventilation that results in insufficient dilution or removal of infectious droplet nuclei
Air circulation	Recirculation of air containing infectious droplet nuclei
Specimen handling	Improper specimen handling procedures that generate infectious droplet nuclei
Air Pressure	Positive air pressure in infectious patient's room that causes <i>M. tuberculosis</i> organisms to flow to other areas

Factors related to transmission: Exposure

Table 2.4
Proximity and Length of Exposure Factors that Can Affect
Transmission of *M. tuberculosis*

Factor	Description
Duration of exposure to a person with infectious TB	The longer the duration of exposure, the higher the risk for transmission
Frequency of exposure to infectious person	The more frequent the exposure, the higher the risk for transmission
Physical proximity to infectious person	The closer the proximity, the higher the risk for transmission

Diagnosis of active pulmonary TB

Chest X-
ray

Fast and sensitive

Sputum

tests

- Sputum smear : Acid-fast stain is easy and quick
- select highly infectious patients,
- but it does not confirm a diagnosis of TB
- Sputum culture : gold standard, but takes 1 - 2 months
- Nucleic acid amplification testing(NAA) : Fast, but cannot replace culture

Aims of TB Treatment

- Cure the patient of TB
- Prevent death from active TB or its latent effects
- Prevent relapse of TB
- Decrease transmission of TB to others
 - Infectiousness decreased by 90% after 2 days of treatment
 - Infectiousness decreased by 99% after 7 days of treatment
- Prevent the development of acquired resistance

肺結核都治計畫(DOTS)

- **Cure is the best prevention !**
- Directly Observed Therapy Shortcourse
- Health care worker watches patient swallow each dose of medication (Every pill, every day)
- 所謂都治計畫(DOTS) ，是指醫護人員或經過訓練的關懷員，每日親自關懷並目視病人服下每一顆應服的藥，以提高治癒率。這種觀察治療的方式是世界衛生組織認為目前最有效的肺結核防治策略。

送藥到手

服藥入口

吃完再走

表 4、各單位空氣中結核菌 SDA 之陽性率(以總測量次數計)

單位	暴露區 高			低			P-value
	測量數	陽性數	%	單位	測量數	陽性數	%
胸腔科病房	12	2	16.67	行政單位	8	0	0.00
呼吸加護病房	6	0	0.00	精神科病房	4	1	25.00
檢驗科	6	0	0.00	婦產科病房	4	0	0.00
內科加護病房	6	1	16.67	小兒科病房	8	1	12.50
病理科	6	0	0.00	藥局	4	0	0.00
外科病房(二期)	12	1	8.33	病歷室	4	1	25.00
門診	6	1	16.67	彰美診所	4	1	25.00
肺功能及支氣管鏡室	6	1	16.67				
急診觀察室(ACU)	6	1	16.67				
急診(ER)	6	0	0.00				
台中醫院門診	4	0	0.00				
台中醫院病房	8	2	25.00				
總數	84	9	10.71	總數	36	4	11.11
							1.00

附註：同一單位同一時間四個採樣結果中任一陽性則定義該時點為陽性

表 5、醫院員工 PPD 轉陽性率的比較

暴露區		高		低				P-
		PPD 檢		PPD 檢				
單位	驗數	陽性數	%	單位	驗數	陽性數	%	value
胸腔科病房	6	3	50.00	行政單位	10	2	20.00	
呼吸加護病房	3	1	33.33	精神科病房	1	0	0.00	
檢驗科	11	5	45.45	小兒科病房	4	0	0.00	
內科加護病房	8	2	25.00	藥局	12	0	0.00	
病理科(院牧部)	4	1	25.00	病歷室	17	3	17.65	
外科病房(二期)	4	2	50.00					
門診	11	3	27.27					
急診(ER)	8	0	0.00					
Total	55	17	30.91	Total	44	5	11.36	0.028

醫療機構內的結核病傳播

■ 發生的危險性取決於下列因素

- 醫療機構內的結核病盛行率
- 社區內的結核病盛行率
- 服務的病人族群
- 特定科別或進行特定檢查
 - 例如支氣管鏡、插管、抽痰、引發飛沫產生之檢查、解剖等，工作時與上述檢查有近距離接觸者受傳染的危險性較大。
- 院內感染管制措施執行的效率



造成院內感染原因

- 罹患肺結核或喉部結核的病患延遲診斷或沒有被診斷出來
 - 可能原因：非典型病例，醫師對肺結核病人之懷疑警覺過低，實驗室報告延誤等
- 肺結核病人沒有合適的隔離
 - 可能原因：隔離病房不足，隔離病房未符合標準，病人沒有遵照隔離原則而擅自離開病房等
- 肺結核病人沒有接受適當的抗結核菌藥物治療
 - 可能原因：藥物敏感性試驗報告延遲，醫師未依照結核病治療指引的建議，未進行都治計畫等
- 未落實感染控制措施



感染控制計畫

■ 第一優先：行政管理

- 可防止或降低帶有結核菌飛沫產生以減低健康照護人員 (Health-care worker, HCW) 與病人暴露機會

- 早期診斷可能的感染者
- 即時隔離有傳染性的病人
- 儘快適當的治療
- 評估院內傳播結核菌的危險程度
- 制定感染控制計畫
 - 書面感控SOP、防護措施、HCW教育訓練
- 指派專職人員，賦與權責監督計畫實施



■ 第二優先：環境控制

- 在高危險區域採取環境的防護措施，以減少空氣中傳染性飛沫的濃度
- 利用自然通風、控制空氣流動

■ 第三優先：個人防護

- 使用可遮蔽口鼻，並可濾除含有結核菌飛沫的呼吸道防護
- N95口罩

SARS



認識嚴重急性呼吸道症候群

SARS

傳染途徑 經由飛沫或透過空氣

症狀 發燒、乾咳、頭痛、肌肉痛及上呼吸道感染症狀

感染高危險群 在本(92)年2月1日之後曾到過有SARS報告病例的地區、或曾照顧SARS個案或與其共同生活、於上述經歷後10天內出現38℃以上高燒及伴有咳嗽、呼吸急促或呼吸困難等一種以上的呼吸道症狀者，請儘速就醫。

預防措施

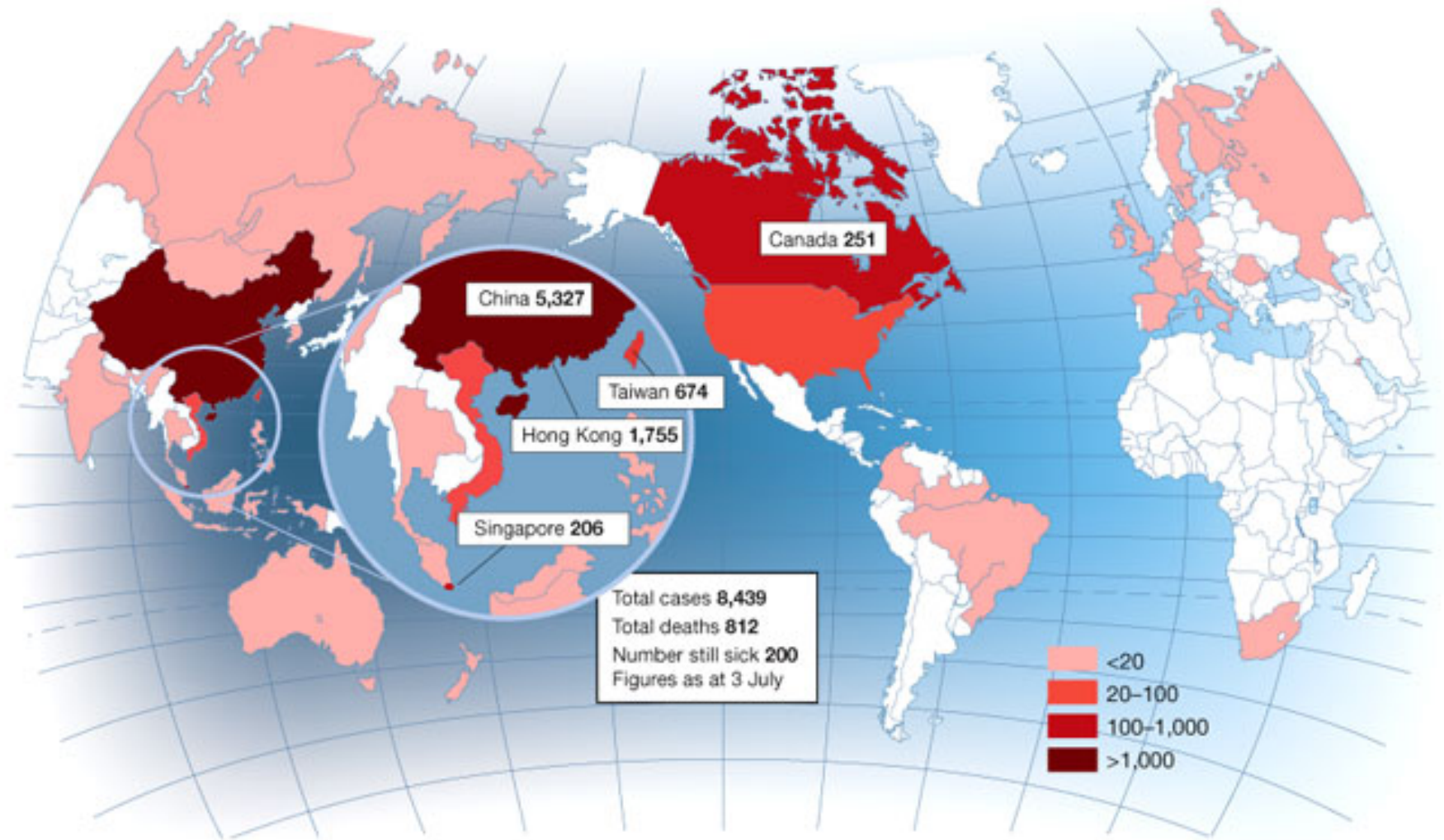
- 1 勤洗手
- 2 保持環境衛生及空氣流通
- 3 避免到人群聚集或空氣不流通的地方
- 4 避免不必要的探病
- 5 均衡飲食
- 6 適量休息及運動

衛生署疾病管制局 關心您

疫情通報專線 0800-024582

疾病管制局全球資訊網 www.cdc.gov.tw

Total number of SARS cases worldwide



Affected 26 countries

SARS-CoV: Transmission

- The primary way that SARS appears to spread is by **close person-to-person contact**.
- Droplet spread can happen when droplets from the cough or sneeze of an infected person are propelled a short distance (generally up to 3 feet) through the air and deposited on the mucous membranes of the mouth, nose, or eyes of persons who are nearby.
- The virus also can spread when a person touches a surface or object contaminated with infectious droplets and then touches his or her mouth, nose, or eye(s).
- In addition, it is possible that SARS-CoV might be spread more broadly through the air (airborne spread).

SARS and healthcare workers

- Healthcare workers (HCWs) were at high risk and accounted for a **fifth** of all cases globally
- Risk factors :
 - Lack of awareness and preparedness when the disease first struck
 - Poor institutional infection control measures, lack of training in infection control procedures
 - Poor compliance with the use of personal protection equipment (PPE)
 - Exposure to high-risk procedures such as intubation and nebulization
 - Exposure to unsuspected SARS patients.

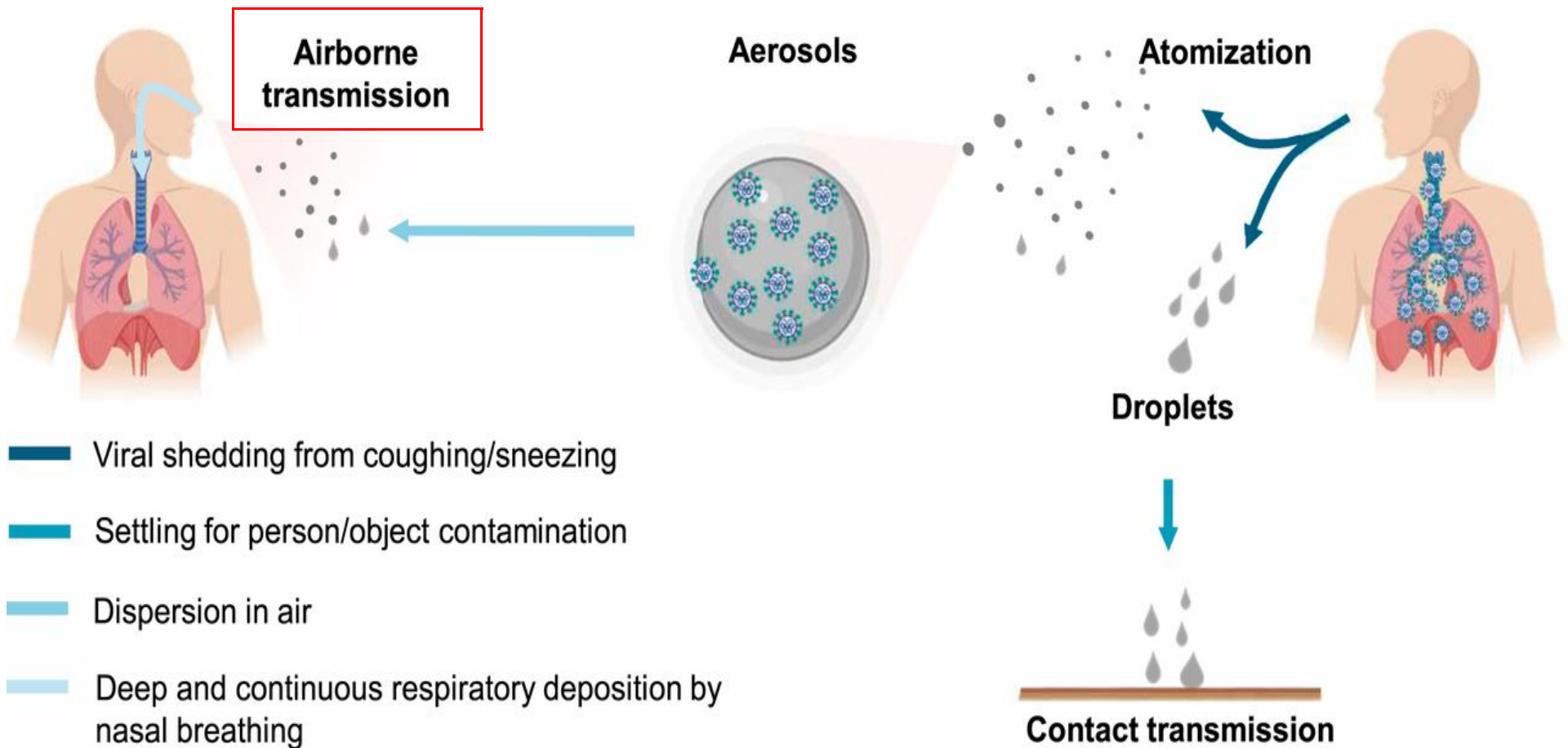
COVID-19

TABLE 4

Estimates of the numbers of deaths in HCWs due to COVID-19 (January 2020–May 2021) using various methods

Scope	HCWs deaths reported to the WHO COVID-19 Surveillance	A summary of estimations		
		Value/range	Method	Comments
Global (HCWs)	6643	115 500	Population-based estimate	
		83 000 to 133 000	Triangulation A: Using Indirect standardization (by sex and age)	Upper range adjusts for the 60% under-estimation of deaths
		80 000 to 160 000	Triangulation B: Using results of meta-analysis (based on PCR testing)	
		179 500	Population-based estimate using IHME estimated overall deaths due to COVID-19	See Annex 4 (Table 1)

COVID-19 transmission



COVID-19

•Routes of Transmission

- Droplet transmission
- Contact with contaminated surfaces
- Airborne transmission (aerosols) > 6 feet (*poor ventilation, >15 minutes exposure, heavy breathing*)

•Incubation Period

- 2–14 days; average 5–6 days for original strain; omicron shortened to 2–4 days

•Infectivity:

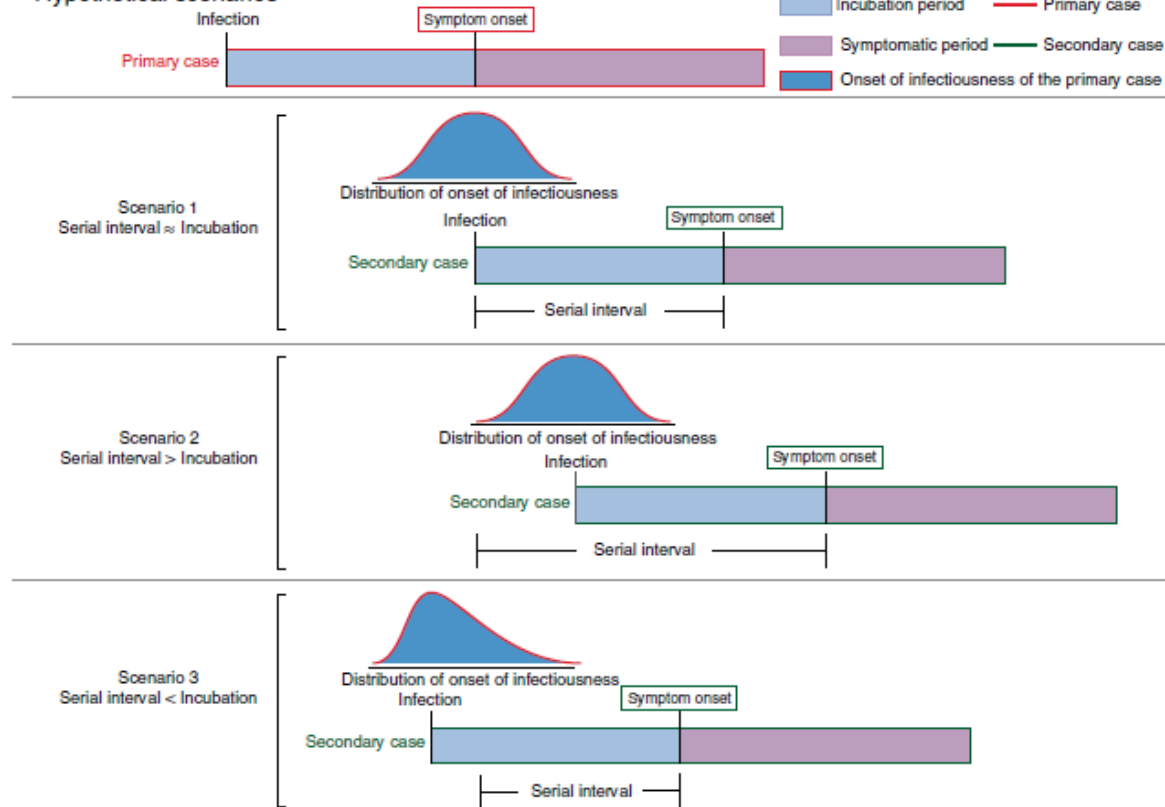
- Presymptomatic individuals are infectious **1–3 days before symptom onset**
- Asymptomatic infection rate: **24%–38%** (*PLOS Med.* 2020;17(9):e1003346)
- **Asymptomatic and presymptomatic individuals account for >50% of new infections** (*JAMA Netw Open.* 2021;4(1):e2035057)

Reasons for healthcare workers becoming infected with novel coronavirus disease 2019 (COVID-19) in China

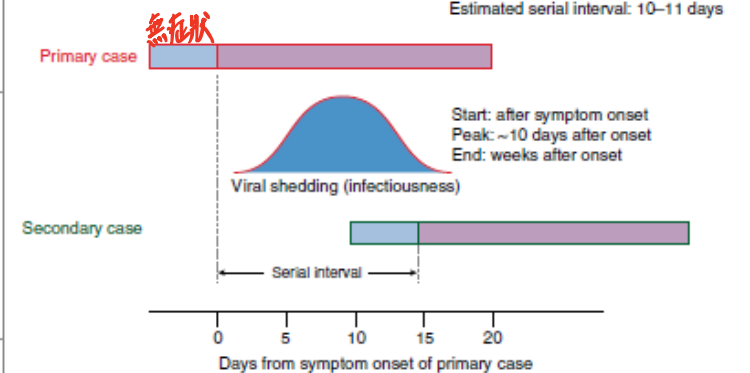
- Up until February 24th, 2055 healthcare workers (community/hospital-acquired not to be defined) had been confirmed infected with COVID-19, with 22 (1.1%) deaths.
- Reasons:
 1. Inadequate personal protection of healthcare workers at the beginning of the epidemic (did not understand the pathogen well)
 2. Long-time exposure to large numbers of infected patients, pressure of treatment, work intensity, and lack of rest
 3. Shortage of personal protective equipment (PPE)
 4. Inadequate training for IPC, leaving them with a lack of knowledge of IPC for respiratory-borne infectious diseases

a Covid

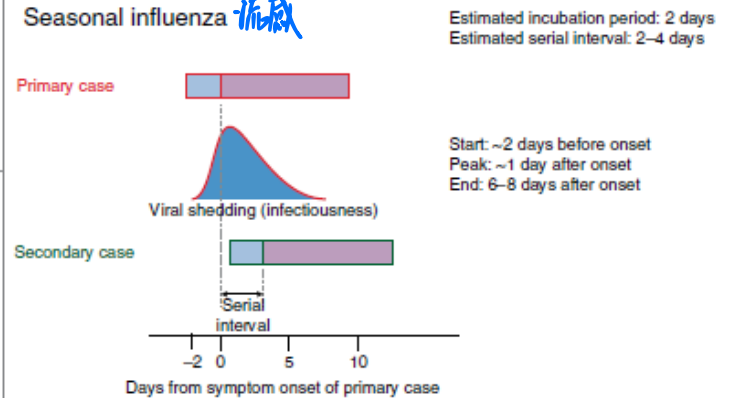
Hypothetical scenarios



SARS 2003



Seasonal influenza 流感

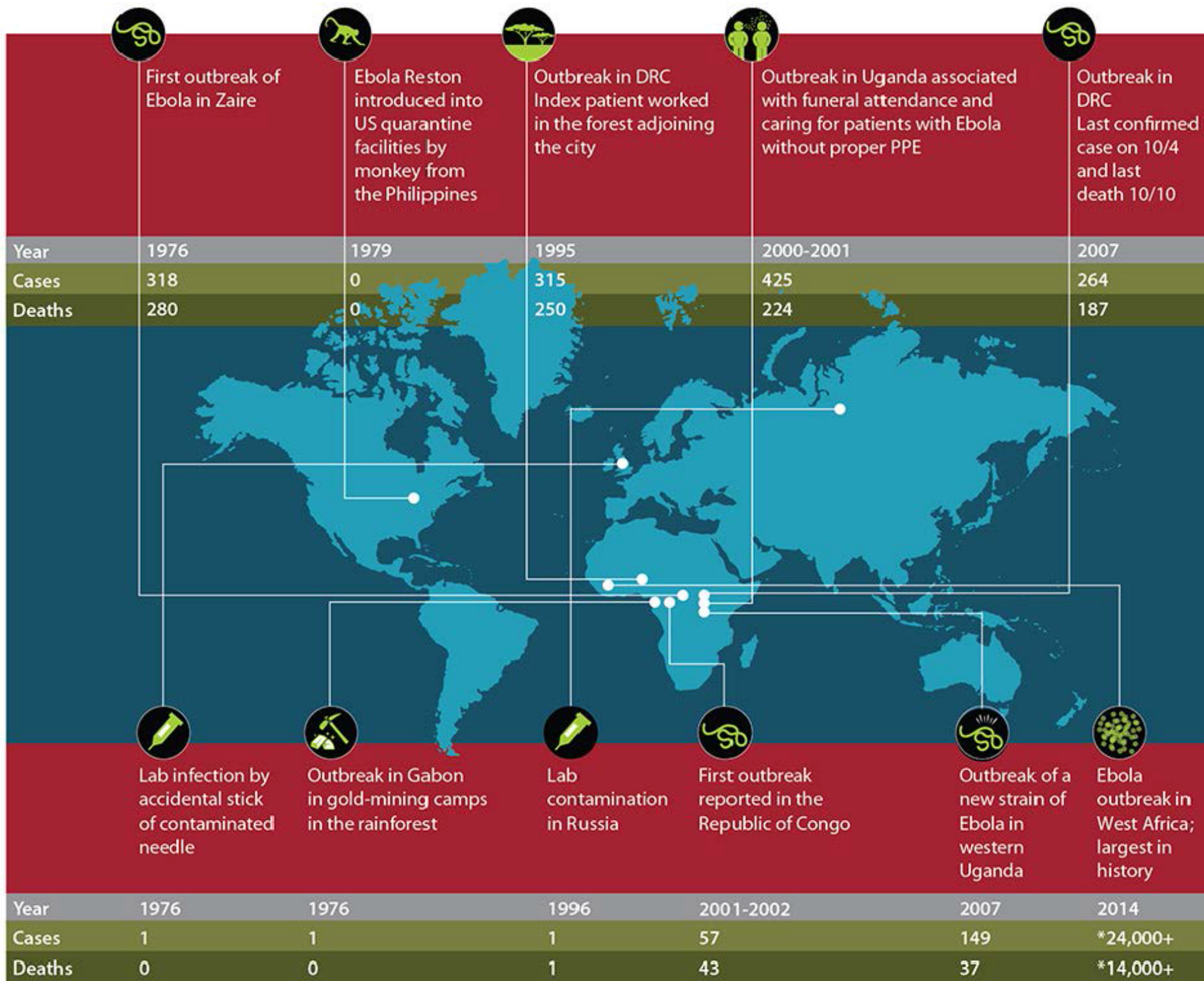


b ID

It was estimated that 44% of secondary COVID-19 cases were transmitted by people who had not yet developed symptoms.

Ebola





Mortality in 2014 Ebola outbreak

	Guinea	Liberia	Sierra Leone
General population	0·02%	0·11%	0·06%
Healthcare workers	1·45%	8·07%	6·85%





USAID
FROM THE AMERICAN PEOPLE

DEMOCRATIC REPUBLIC OF THE CONGO – EBOLA OUTBREAK

FACT SHEET #6, FISCAL YEAR (FY) 2020

2018–2020

MAY 19, 2020

NUMBERS AT A GLANCE

3,462

Total Confirmed and
Probable EVD Cases
MoH – May 19, 2020

2,279

Total EVD-Related
Deaths
MoH – May 19, 2020

0

New Confirmed Cases in
Past 21 Days
MoH – May 19, 2020

HIGHLIGHTS

- New EVD cluster identified in Beni after more than 50 days without cases; response actors augment activities to contain EVD spread
- EVD-positive individual flees health facility, prompting concerns of additional transmission
- USAID/OFDA provides nearly \$23.4 million in additional funding for EVD response activities in the DRC

HUMANITARIAN FUNDING

FOR THE DRC EBOLA OUTBREAK RESPONSE IN
FYs 2018–2020

USAID/OFDA ¹	\$276,782,247
USAID/FFP ²	\$36,975,000
USAID/GH ³	\$11,979,389
USAID in Neighboring Countries	\$16,671,381
\$342,408,017⁴	

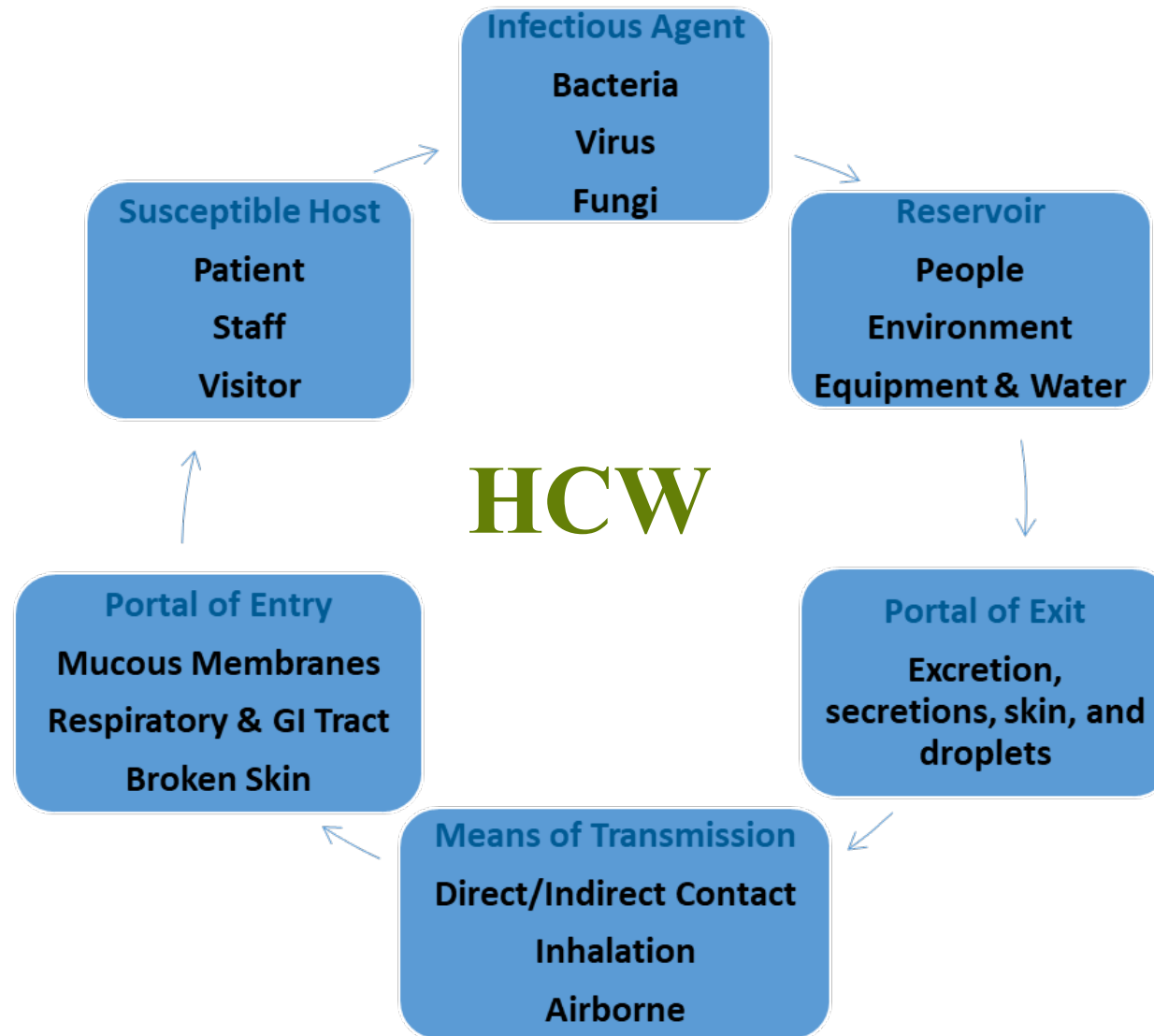
KEY DEVELOPMENTS

- The Government of the Democratic Republic of the Congo (GoDRC) Ministry of Health (MoH) recorded a new confirmed Ebola virus disease (EVD) case in North Kivu Province's Beni Health Zone on April 10, the first confirmed case detected in 52 days. From April 11 to 27, the MoH recorded six additional confirmed EVD cases in Beni.

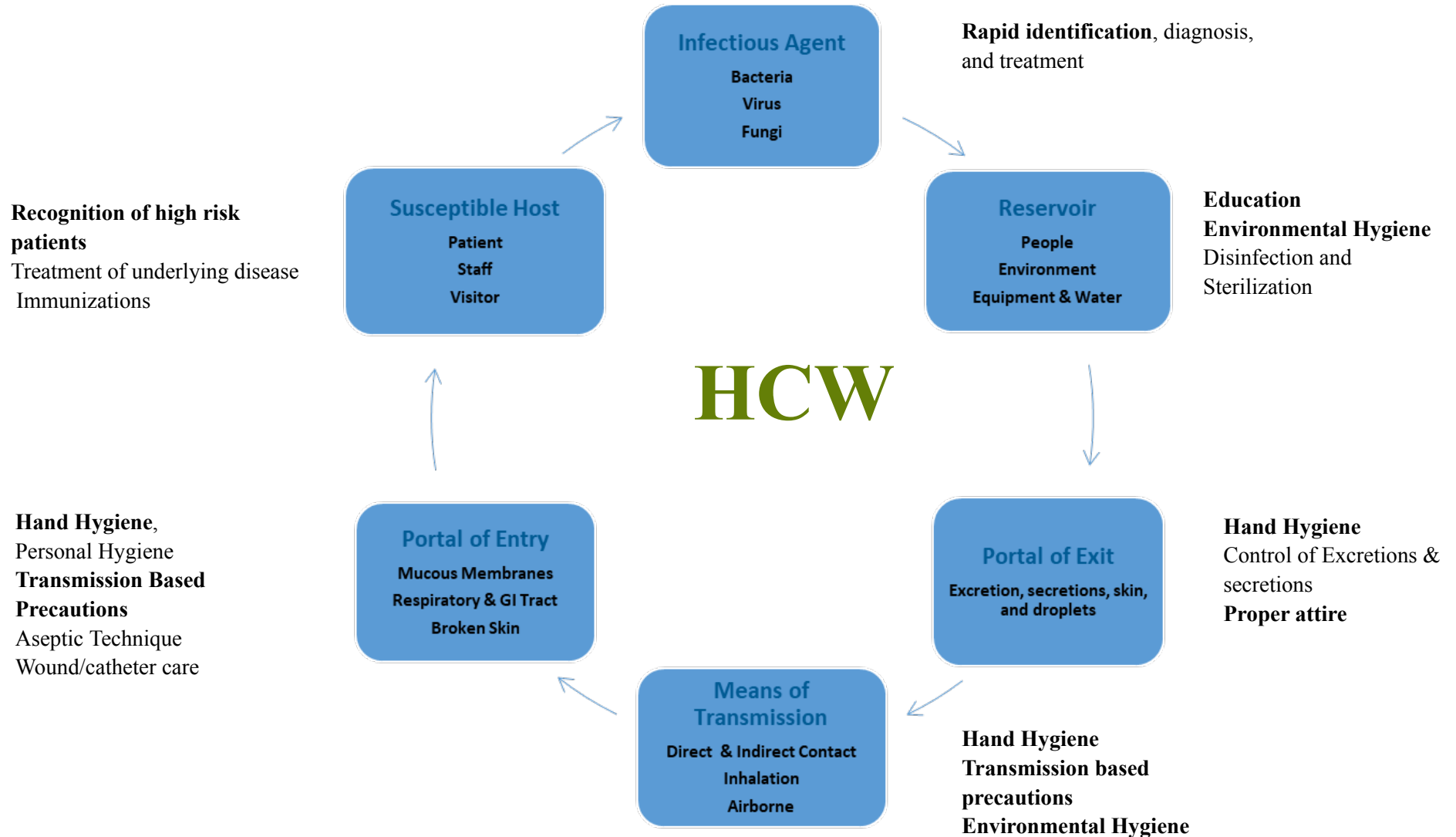
Outline

- Introduction to Occupational Infectious diseases
- Occupational health hazards in health care workers
- Needle-stick injury(針扎) – hepatitis B virus 、 hepatitis C virus 、 HIV/AIDS
- Respiratory infection - tuberculosis
- SARS, COVID-19
- Ebola
- Standard precautions in health care

The Chain of Infection



Breaking the chain of infection



Three basic routes of transmission

- Contact

- Direct
- Indirect



- Droplet

- Larger; don't travel long distances, not infective over time
- Spatial separation (≥ 3 feet)

- Airborne

- Smaller; infective over time and distance



Standard Precautions

- Standard precautions are the basic level of infection control that should be used in the care of **all patients all of the time**, regardless of suspected or confirmed infection status.
- Use standard precautions in the care of all patients to reduce the risk of transmission of microorganisms from both recognized and non-recognized sources of infection.
- Applies to blood, all body fluids, secretions and excretions (except sweat) whether or not they contain visible blood; non-intact skin; and mucous membranes.

Standard Precautions

Personal Protective Equipment (PPE) for Standard Precautions

- Gloves
- Gowns 隔離衣
- Masks and Respirators
- Face and Eye Protection

Additional Procedures

- Hand Hygiene
- Patient Care Equipment
- Injection Safety

Hand Hygiene

- Hand Hygiene (HH) has been cited frequently as the **single most important practice to reduce the transmission of infectious agents in healthcare settings** and is an essential element of Standard Precautions.
- **The term “hand hygiene” includes:**
 - Handwashing with soap and water
 - Use of alcohol-based products that do not require the use of water (**>70% alcohol**)
 - No artificial nails for those having contact with (high-risk) patients

Prevention of the Transmission of Disease

- Health care facility's infection control program
- Immunization of health care workers
- Training of health care personnel
- Policies and procedures for device- and procedure-oriented services
- Reporting and tracking of the transmission of disease by the CDC
- Provision of personal protective equipment (PPE) such as gloves, gowns, lab coats, masks, and face shields in appropriate sizes and in accessible locations

Prevention of the Transmission of Disease

- Provision of adequate hand-washing facilities and supplies
- Identification of jobs with high-risk exposure
- Provision of medical care and evaluation for any employee who has an exposure incident
- Provision of appropriate waste and disposal containers and guidelines for disposal
- Enforcement of rules of no eating, drinking, smoking, handling contact lenses, and mouth-pipetting or suctioning in any area that can be potentially contaminated.

